Act as a licensed PE an...
TXT 5.82KB

Thought for 4 seconds

The user has provided a detailed prompt for a comprehensive exam-focused review of formwork design for the NCEES PE Civil: Construction exam. They've also supplied a file containing this prompt. The task is to generate a thorough review that aligns with the NCEES specifications and the PE Civil Reference Handbook, Version 2.0.

I need to search for relevant technical content to ensure accuracy and completeness. I'll search for information on ACI 347R lateral pressure formulas, formwork load combinations, shoring and reshoring practices, and OSHA requirements. I'll also look for NCEES PE Civil exam specifications and the PE Civil Reference Handbook details. These searches will help me gather the necessary technical data and standards to produce a well-structured, exam-focused review.

Found 57 web pages

The search results provide some relevant information. I need to open several of these results to gather detailed content for the review. I'll open the ACI 347R lateral pressure formulas result, the formwork load combinations result, the shoring/reshoring result, the OSHA requirements result, and the NCEES exam specifications result.

Read 5 pages

For purposes of applying the formulas, ACI 347 defines a wall as a vertical element with at least one plan dimensi... ↗

The design load for falsework shall consist of the sum of dead and live vertical loads, and a design horizontal load ↗

Table 5.5 — Simplified Analysis of Loads on Shores and Slabs of Multistory Structure ↗

1926.703 - Requirements for cast-in-place concrete. - - Part Number:1926 ↗

NCEES Principles and Practice of Engineering Examination CIVIL-CONSTRUCTION CBT Exam Specifications ↗

I need to gather more specific information on lateral pressure formulas and coefficients. I'll search for "ACI 347R Cw Cc coefficients" and "concrete formwork lateral pressure formula psi". I also need details on formwork component design, bracing requirements, form removal criteria, and common exam traps. I'll search for those as well.

Found 42 web pages

I have gathered information from multiple sources. Now I need to structure the comprehensive review. I'll organize it into sections covering loads, lateral pressure, components, shoring, bracing, removal, applications, safety, and common errors. I'll include formulas, step-by-step procedures, worked examples, practice problems, and a high-yield checklist. I'll cite the sources appropriately. The user's prompt also mentions

> specific topics like minimum combined vertical design loads (100 psf, 125 psf), lateral pressure formulas, coefficients Cw and Cc, hydrostatic cap, minimum design pressure, form ties, shoring/reshoring, bracing, form removal, and OSHA requirements. I need to ensure all these are covered. I'll also flag any details not specified by NCEES. The answer should be exam-focused and suitable for rapid review. I'll now begin writing the comprehensive response.

Formwork Design for the NCEES PE Civil: Construction Exam

Exam-Focused Review | Design for Support of Construction Loads

Scope Note: This review covers only **Formwork** within the "Design for Support of Construction Loads" area. Falsework, scaffolding, and support of excavation are referenced only where necessary to explain formwork behavior. All content is aligned with the NCEES PE Civil: Construction CBT Exam Specifications (effective April 2024) and the PE Civil Reference Handbook, Version 2.0.

1. Formwork Loads and Load Combinations

1.1 Definitions and Load Categories

Formwork must be designed to resist all vertical and lateral loads that may reasonably be anticipated during construction. OSHA 1926.703(a)(1) mandates this explicitly and deems conformance with the appendix to §1926.703(a)(1) as meeting the requirement [30] .

Load Type	Components	Typical Magnitude / Notes
Vertical dead loads	Weight of concrete, reinforcement, formwork, falsework, embedded items	Concrete: 150 pcf normal-weight; 160 pcf for bridge falsework (with reinforcement) [11]
Vertical live loads	Personnel, equipment, material storage, impact	Minimum 50 psf per ACI 347R; 75 psf for motorized carts; 100 psf combined dead + live minimum
Lateral loads	Fresh concrete pressure, wind, seismic, equipment, construction	See Section 2 for concrete pressure

Load Type	Components	Typical Magnitude / Notes
	sequence	
Horizontal construction loads	Equipment, sidesway, eccentric loading, wind	Minimum 3% of total dead load for falsework bracing (WSDOT) [11]

1.2 Minimum Combined Vertical Design Loads

ACI 347R specifies minimum design loads for combined dead and live loads:

- **100 psf** – Minimum for combined dead and live loads regardless of structure thickness
- **125 psf** – Minimum when using buggies to transport concrete

These minimums apply to the total design load for formwork and falsework. The 100 psf minimum is a recurring exam trap: it applies **regardless of structure thickness**, not just for thin slabs.

Live load allowance within the minimum: For falsework design (WSDOT), live loads consist of actual equipment loads applied as concentrated loads plus a minimum uniform load of **25 psf** over the entire plan area, plus **75 lb/ft** at deck overhangs [11] . This is falsework-specific but illustrates the distinction between uniform construction live load and concentrated equipment loads.

1.3 Load Combinations for Formwork Design

Formwork design typically uses **allowable stress design (ASD)** with the following conceptual combinations:

1. **Dead + Live (vertical):** Concrete + reinforcement + formwork + construction live load (minimum 50 psf, or 75 psf for motorized equipment)
2. **Dead + Lateral concrete pressure:** For vertical forms — concrete weight and pressure govern separately
3. **Dead + Wind:** Minimum 15 psf wind for formwork bracing (see Section 6)
4. **Dead + Construction horizontal:** Minimum 100 lb/ft at slab edge or 2% of vertical load (see Section 6)

Critical exam note: ACI 347R does **not** use LRFD load factors. Formwork design is ASD-based with allowable stresses and safety factors built into component capacities.

1.4 Common Errors and Exam Traps – Loads

Error	Correct Approach
Using 50 psf live load when 125 psf required	Check if motorized carts/buggies are used → 125 psf minimum
Forgetting the 100 psf combined minimum	Apply 100 psf floor even for thin slabs
Applying LRFD factors to formwork	Formwork uses ASD; use service loads and allowable stresses
Neglecting impact from concrete placement	Include impact allowance in live load
Confusing falsework vs. formwork live loads	Formwork: 50 psf minimum; falsework: 25 psf uniform + concentrated equipment

2. Lateral Pressure of Fresh Concrete

2.1 Factors Affecting Lateral Pressure

Lateral pressure on vertical formwork depends on:

- **Rate of placement (R):** Higher pour rate → higher pressure
- **Concrete temperature (T):** Lower temperature → slower set → higher pressure
- **Placement height (h):** Hydrostatic cap = 150h psf
- **Mix chemistry (Cc):** Retarders increase pressure
- **Unit weight (Cw):** Lightweight concrete reduces pressure
- **Consolidation method:** Internal vibration increases pressure

2.2 ACI 347R Lateral Pressure Formulas

Wall Forms (ACI 347R-14):

$$P_{\max} = C_w C_c \left(7.2 + \frac{720R}{T + 17.8}\right)$$

Column Forms (ACI 347R-14):

$$P_{\max} = C_w C_c \left(7.2 + \frac{785R}{T + 17.8}\right)$$

Where:

- $P_{\max}$ = lateral pressure, psf
- R = rate of placement, ft/hr
- T = concrete temperature during placement, °F
- C_w = unit weight coefficient
- C_c = chemistry coefficient

Key distinctions:

- **Wall:** at least one plan dimension > 6.5 ft [1]
- **Column:** no plan dimension > 6.5 ft [1]
- Column formula uses **785** instead of **720** — columns are confined on four sides and poured more rapidly

2.3 Coefficients Cw and Cc

Unit Weight Coefficient (Cw):

$$C_w = 0.5\left(1 + \frac{w}{145}\right) \quad (\text{but not less than } 0.80)$$

For normal-weight concrete (w = 145 lb/ft³): Cw = 1.0

Chemistry Coefficient (Cc):

Admixture Type	Cc
No admixture	1.0

Admixture Type	Cc
Lignosulfonate retarder	1.0
Naphthalene-based superplasticizer	1.2
Polycarboxylate (high-range water reducer)	1.4
Blends with retarder	1.5

Exam trap: Cc = 1.0 for no admixture AND for lignosulfonate retarder. Many candidates incorrectly assign 1.2 to all retarders.

2.4 Hydrostatic Cap and Minimum Pressure

Hydrostatic cap:

$$P_{\text{hydrostatic}} = 150h \quad (\text{psf})$$

The formula-derived pressure shall **never exceed** 150h. If the form is filled to full height in less than the time required for concrete to begin to stiffen, use $p = wh$ directly [1].

Minimum design pressure:

$$P_{\min} = 600C_w \quad (\text{psf})$$

The formula-derived pressure shall **never be less than** 600Cw psf.

Maximum limits: The formulas do not apply when using $p = wh$. For bottom-up pumping, use wh plus at least 25% for pump surge pressure [1].

2.5 Design Limits Summary

Condition	Governing Pressure
Formula result > 150h	Use 150h (hydrostatic governs)
Formula result < 600Cw	Use 600Cw (minimum governs)

Condition	Governing Pressure
Bottom-up pumping	wh + 25% surge
Full-height rapid fill	wh (hydrostatic)

2.6 Worked Example – Wall Form Pressure

Problem: A wall form is 12 ft high. Concrete is placed at 6 ft/hr at 70°F. Normal-weight concrete, no admixtures. What is the maximum lateral pressure?

Solution:

1. Cw = 1.0, Cc = 1.0
2. $P = 1.0 \times 1.0 \times (7.2 + 720 \times 6/(70 + 17.8))$
3. $P = 7.2 + 4320/87.8 = 7.2 + 49.2 = 56.4$ … wait, this is clearly wrong dimensionally. The formula is in psf with R in ft/hr and T in °F. Let me recalculate:

Actually, the ACI 347R formula in USCS units is:

$$P_{\max} = C_w C_c \left(150 + \frac{9000 R}{T}\right)$$

This is the older form. The SI form uses 7.2 and 720/785. For USCS:

$$P_{\max} = C_w C_c \left(150 + \frac{9000 R}{T}\right)$$

Let me verify: The search result shows "P = 150 + 9,000×R / T with a maximum of 3… a minimum of 600 lbs per sq ft" 🔗 . Yes, the USCS form is:

$$P = 150 + \frac{9000 R}{T}$$

with minimum 600 psf and maximum 150h.

Recalculated:

1. $P = 1.0 \times 1.0 \times (150 + 9000 \times 6/70)$

2. $P = 150 + 54000/70 = 150 + 771.4 = 921.4$ psf
3. Check minimum: 921.4 > 600 ✓
4. Check hydrostatic cap: 150 × 12 = 1800 psf; 921.4 < 1800 ✓
5. **Answer: 921 psf**

Column form example: Same conditions but column form uses:

$$P = C_w C_c \left(150 + \frac{9000R}{T}\right) \quad ?$$

Actually, the column formula has a different constant. The SI form uses 785 vs 720. The USCS equivalent for columns is:

$$P = 150 + \frac{9000R}{T} \quad (\text{wall})$$

$$P = 150 + \frac{9825R}{T} \quad (\text{column})?$$

This needs verification. The search results are inconclusive for the exact USCS column constant. **Flag:** The exact USCS column constant should be verified against ACI 347R-14 directly. The SI forms clearly show 720 (wall) and 785 (column) .

2.7 Common Errors – Lateral Pressure

Error	Consequence
Using hydrostatic when formula governs	Over-design; but using formula when hydrostatic governs = under-design
Forgetting 600 psf minimum	Under-design for slow pours
Confusing wall (720) and column (785) constants	Wrong pressure
Neglecting Cw for lightweight concrete	Over-design
Using Cc = 1.2 for all retarders	Over-design (lignosulfonate = 1.0)

Error	Consequence
Forgetting 25% pump surge for bottom-up pumping	Under-design
Using T in °C with °F formula	Gross error

3. Formwork Components and Structural Design

3.1 Component Hierarchy

Component	Function	Typical Material
Sheathing	Direct contact with concrete; distributes pressure	Plywood, lumber, steel
Studs	Support sheathing; vertical (walls) or horizontal (slabs)	Lumber, steel
Wales	Support studs; horizontal (walls) or vertical	Lumber, steel, aluminum
Form ties	Resist lateral pressure; hold wall forms together	Steel rods, coil ties
Shores	Vertical support for elevated forms	Steel posts, lumber
Braces	Lateral stability and alignment	Lumber, steel, turnbuckles
Hardware	Wedges, clamps, brackets	Steel

3.2 Tributary Area and Load Distribution

Each formwork component supports loads from a **tributary area**. For form ties:

$$F_{\text{tie}} = P_{\max} \times A_{\text{trib}}$$

Where A_{trib} = tie spacing (horizontal) × tie spacing (vertical).

Example: Ties at 3 ft × 3 ft = 9 ft² tributary area. If Pmax = 1,168.5 psf, tie tension = 9 × 1,168.5 = 10,517 lb .

3.3 Bending, Shear, and Deflection Checks

Formwork components are designed as beams:

Bending:

$$M = \frac{wL^2}{8} \quad \text{(simple span, uniform load)}$$

$$f_b = \frac{M}{S} \leq F_b'$$

Shear:

$$V = \frac{wL}{2}$$

$$f_v = \frac{1.5V}{A} \leq F_v'$$

Deflection:

$$\Delta = \frac{5wL^4}{384EI} \leq \Delta_{\text{allow}}$$

Deflection limits (ACI 347R):

- Sheathing: l/360
- Studs, wales, ties: l/240
- Shores: l/240

Allowable stresses: Use NDS for wood, AISC for steel. Apply appropriate adjustment factors (CD for load duration, etc.).

3.4 Form Tie Design

Form ties must be designed for the full tributary lateral pressure. Key provisions:

- Minimum working strength: **3,000 lb** when fully assembled (some specifications)
- Must be adjustable in length to permit tightening
- Must leave no metal closer than **1 inch** from concrete surface (or 1.5 inch for some specs)
- Water-seal type required for water-holding structures

3.5 Common Errors – Component Design

Error	Correct Approach
Forgetting tributary area	F_tie = P × (s_h × s_v)
Using wrong deflection limit	Sheathing: l/360; studs/wales: l/240
Ignoring load duration factor (CD) for wood	Wood formwork is temporary → CD = 1.25 for construction loads (≤7 days)
Not checking bearing stress at supports	Verify crushing at stud/wale connections
Using gross section for steel ties	Use net area for threaded rods

4. Shoring and Reshoring

4.1 Purpose and Load Path

Shoring: Temporary vertical supports that carry the weight of fresh concrete and formwork until the concrete gains sufficient strength.

Reshoring: Supports installed after formwork removal to distribute loads from newly placed concrete to multiple floors below.

Load path: Fresh concrete + formwork → shores → floor below → reshores → multiple floors → foundation.

4.2 Multistory Construction Load Distribution

ACI 347.2R provides simplified load distribution tables. For a two-level shoring, one-level reshoring system:

Step	Operation	Fresh Slab Load	Hardened Slab Load	Shore Load
1	Place Level 1 concrete	1D	0	1D
2	Place Level 2 concrete	2D	0	2D
3	Remove first-level shores	1D	1D	0
4	Place reshores (snug)	1D	1D	0
5	Place Level 3 concrete	2D	1D	1D
6	Remove first-floor reshores	1.33D	1.33D	0.67D [20]

Maximum shore/reshore load: 1.5D or 169 psf for both three- and two-reshore systems .

4.3 Simplified Load Distribution (Exam-Critical)

Rule of thumb for reshoring:

- Top reshore carries **2/3** of the load
- Lower reshore/slab carries **1/3** of the load

100% reshoring plan: 8 reshores for every 10 shores removed .

4.4 OSHA Written Design Requirement

OSHA 1926.703(b)(8)(i): Whenever single post shores are used one on top of another (tiered), the design of the shoring shall be prepared by a **qualified designer** and the erected shoring shall be inspected by an **engineer qualified in structural design** [30] .

This is a high-yield exam trap: The written design requirement applies to **tiered single post shores**, not all shoring. However, drawings or plans for formwork (including shoring equipment) must be available at the jobsite at all times [30] .

4.5 Worked Example – Shore Load

Problem: A one-way slab is 8 in. thick (D = 100 psf). Two levels of shoring, one level of reshoring. What is the maximum shore load during Level 3 placement?

Solution:

- D = 100 psf (8/12 × 150)
- Maximum shore load = 1.5D = 150 psf
- This occurs when placing Level 3 concrete
- **Answer: 150 psf**

4.6 Common Errors – Shoring

Error	Correct Approach
Assuming all shore loads = 1D	Maximum is 1.5D for 2-shore/1-reshore system
Forgetting load path through reshores	Reshores distribute load to multiple floors
Missing OSHA written design for tiered shores	Required for single post shores stacked
Using shore capacity without checking slab capacity	Slab must be checked for construction loads
Neglecting precompression of reshores	Reshores must be snug but not loaded initially

5. Bracing and Anchorage

5.1 Lateral Load Requirements

Minimum lateral load at each floor line:

$$H = \max(100\ \text{lb/ft of floor edge},\ 2\%\ \text{of total dead load})$$

This applies to shoring and formwork bracing .

Wall form bracing: Minimum **100 lb/ft** of wall .

Minimum wind design load for formwork: 15 psf [11] .

5.2 Bracing for Stability and Alignment

- Bracing must be provided in **two mutually perpendicular directions**
- Diagonal bracing required at each tier of tiered shores [30]
- Braces should be as rigid as possible; wedges required where movement is likely
- Lateral load must be resisted at **each floor line** in both directions

5.3 Common Errors – Bracing

Error	Correct Approach
Using 100 lb/ft only, ignoring 2% dead load	Use the **greater** of the two
Forgetting minimum 15 psf wind	Apply to formwork exposed to wind
Missing bracing in one direction	Two perpendicular directions required
Not bracing tiered shores at splice level	Required for single post shores stacked

6. Form Removal and Reshoring Criteria

6.1 Minimum Time and Strength Requirements

Member Type	Minimum Strength for Form Removal
Walls, columns	Concrete must be adequately set; 2× stress at removal (rule of thumb)
Slabs, beams	70% of design strength (field-cured cylinders)
Structural members	Concrete must support its own weight + any superimposed load

Rule of thumb: Concrete strength at form removal ≥ **2 ×** the stress to which it will be subjected at removal .

ACI 347R: Supporting forms and shores must not be removed from horizontal members until concrete strength is at least **70% of design strength**, as determined by field-cured cylinders or other approved methods .

6.2 Member-Specific Criteria

Member	Typical Removal Time (Type I cement, 70°F)	Strength Requirement
Walls	12–24 hours	2× stress at removal
Columns	24–48 hours	2× stress at removal
Slab soffits (props left)	3–4 days	70% design strength
Beam soffits (props left)	7 days	70% design strength
Props under slabs	14 days	70% design strength
Props under beams	21 days	70% design strength

These are typical values; actual requirements depend on mix design, temperature, and specification.

6.3 Common Errors – Form Removal

Error	Correct Approach
Removing forms without cylinder tests	Field-cured cylinders required for horizontal members
Confusing 2× rule with 70% rule	2× stress = walls/columns; 70% design strength = slabs/beams
Not reshoring when required	Reshore whenever concrete required to support loads in excess of capacity

Error	Correct Approach
Removing props under beams too early	21 days typical for beams; check specification

7. Construction Applications and Quality Control

7.1 Formwork by Element Type

Element	Key Design Considerations
Walls	Lateral pressure governs; form ties critical; bracing for alignment
Columns	Higher pressure formula (785 vs. 720); confined on four sides
Slabs	Vertical loads govern; shoring and reshoring critical
Beams	Combined vertical and lateral; soffit and side forms
Elevated decks	Multistory shoring/reshoring; load distribution
Bridge decks	Higher live loads (160 pcf concrete); wind on falsework towers [11]
Mass concrete	Lower pour rates; temperature control; higher pressure due to slow set

7.2 Inspection and Tolerances

- **Pre-erection:** Inspect all shoring equipment for damage; damaged equipment with reduced strength shall not be used [30]
- **During placement:** Inspect shoring immediately prior to, during, and after concrete placement [30]
- **After placement:** Reinforce any shoring found damaged after erection [30]
- **Tolerances:** Check plumbness, alignment, and elevation of forms

7.3 Documentation and Quality Control

- Drawings or plans for formwork, shoring, working decks, and scaffolds must be available at the jobsite [30]
- Reshoring operations must be documented and inspected
- Concrete maturity and early strength evaluation for form removal decisions

8. Safety and Code Conformance

8.1 OSHA 1926 Subpart Q – Key Formwork Provisions

Section	Requirement
1926.703(a)(1)	Formwork must support all anticipated vertical and lateral loads without failure [30]
1926.703(a)(2)	Drawings/plans available at jobsite [30]
1926.703(b)(1)	Shoring equipment inspected before erection [30]
1926.703(b)(3)	Shoring inspected before, during, after concrete placement [30]
1926.703(b)(8)	Tiered single post shores: written design by qualified designer, inspected by structural engineer [30]
1926.703(b)(10)	Reshoring erected as original forms removed whenever concrete required to support loads in excess of capacity [30]

8.2 ACI 347R Recommended Practice

- Guide to Formwork for Concrete
- Lateral pressure formulas for wall and column forms
- Minimum design loads (100 psf, 125 psf for buggies)
- Deflection limits for formwork components
- Shoring/reshoring load distribution (ACI 347.2R)

8.3 ASCE 37 – Design Loads on Structures During Construction

- Specifies construction loads for temporary structures
- Provides load combinations for construction-stage analysis
- Referenced by NCEES as applicable standard

8.4 Common Errors – Safety and Code

Error	Correct Approach
Not providing drawings at jobsite	Required by OSHA 1926.703(a)(2)
Skipping pre-erection inspection	Required for all shoring equipment
No inspection during placement	Required immediately prior to and during placement
Missing written design for tiered shores	Required by OSHA 1926.703(b)(8)(i)

9. Common Errors and Exam Traps – Master List

#	Trap	Correct Approach
1	Unit inconsistencies (USCS vs. SI)	Check units in every formula; ACI 347R has both
2	Using hydrostatic when formula governs	Compare formula result to 150h; use **lower**
3	Forgetting 600 psf minimum	Apply minimum 600Cw psf
4	Confusing wall (720) and column (785) constants	Wall > 6.5 ft plan dimension; column ≤ 6.5 ft
5	Neglecting Cw and Cc	Cw = 1.0 for normal weight; Cc varies by admixture

#	Trap	Correct Approach
6	Forgetting tributary area for form ties	$F_{tie} = P \times (s_h \times s_v)$
7	Using wrong live load (50 vs. 75 psf)	50 psf minimum; 75 psf for motorized carts
8	Overlooking reshoring load path	Reshores distribute load to multiple floors; max 1.5D
9	Misreading form removal criteria	2× stress = walls/columns; 70% design = slabs/beams
10	Forgetting OSHA written design for tiered shores	Required for single post shores stacked
11	Using LRFD load factors for formwork	Formwork uses ASD
12	Neglecting impact from concrete placement	Include impact allowance
13	Not checking deflection limits	Sheathing l/360; studs/wales l/240
14	Forgetting 25% pump surge for bottom-up pumping	Required by ACI 347R
15	Using T in °C with °F formula	Check temperature units

10. Practice Problems with Detailed Solutions

Problem 1 – Wall Form Lateral Pressure

A wall form is 14 ft high. Concrete is placed at 8 ft/hr at 60°F. Normal-weight concrete (145 pcf) with a naphthalene-based superplasticizer. What is the maximum lateral pressure?

Solution:

1. Cw = 1.0 (normal weight)
2. Cc = 1.2 (naphthalene)
3. $P = 1.0 \times 1.2 \times (150 + 9000 \times 8/60)$
4. $P = 1.2 \times (150 + 1200) = 1.2 \times 1350 = 1620$ psf
5. Check minimum: 1620 > 600 ✓
6. Check hydrostatic cap: 150 × 14 = 2100 psf; 1620 < 2100 ✓
7. **Answer: 1,620 psf**

Problem 2 – Form Tie Design

Using the pressure from Problem 1, design form ties at 2 ft horizontal × 2.5 ft vertical spacing. What is the required tie tension?

Solution:

1. $A_{\text{trib}} = 2 \times 2.5 = 5.0$ ft²
2. $F_{\text{tie}} = 1620 \times 5.0 = 8,100$ lb
3. **Answer: 8,100 lb**

Problem 3 – Shoring Load

A 10-in. thick slab (D = 125 psf) is placed on a two-shore, one-reshore system. What is the maximum shore load during the placement of the floor above?

Solution:

1. D = 125 psf
2. Maximum shore load = 1.5D = 1.5 × 125 = 187.5 psf
3. **Answer: 188 psf**

Problem 4 – Bracing Lateral Load

A floor edge is 100 ft long. Total dead load of the floor = 50,000 lb. What is the minimum lateral bracing load?

Solution:

1. 100 lb/ft × 100 ft = 10,000 lb
2. 2% × 50,000 = 1,000 lb
3. Maximum = 10,000 lb
4. **Answer: 10,000 lb** (100 lb/ft governs)

Problem 5 – Form Removal

A slab is designed for a total load of 150 psf. At form removal, the slab will be subjected to its own weight of 100 psf. What is the minimum concrete strength required?

Solution:

1. Stress at removal = 100 psf
2. Required strength = 2 × 100 = 200 psf
3. As a fraction of design: 200/150 = 1.33 → **133% of design load**?
4. **Correction:** The 2× rule is a rule of thumb for stress, not design load. The actual requirement for horizontal members is **70% of design strength** per ACI 347R .
5. **Answer: 70% of design strength** (governs for slabs)

Note: The 2× stress rule of thumb applies more directly to walls and columns where the stress at removal is compared to the stress the concrete will experience. For slabs and beams, 70% of design strength is the governing criterion.

11. High-Yield Checklist

Loads

□ Minimum combined dead + live = 100 psf (125 psf for buggies)

□ Formwork live load minimum = 50 psf

□ Impact allowance included

□ ASD design (no LRFD factors)

Lateral Pressure

□ Wall vs. column formula (720 vs. 785 in SI)

□ Cw = 1.0 for normal weight

□ Cc = 1.0 for no admixture/lignosulfonate

□ Minimum 600Cw psf

□ Maximum 150h psf

□ 25% pump surge for bottom-up

Components

□ Tributary area for ties

□ Deflection limits (l/360 sheathing, l/240 studs/wales)

□ Bending, shear, deflection checks

□ Wood CD factor for temporary loads

Shoring/Reshoring

□ Maximum shore load = 1.5D

□ Top reshore = 2/3 load; lower = 1/3

□ OSHA written design for tiered shores

□ 100% reshoring = 8 reshores per 10 shores

Bracing

□ 100 lb/ft or 2% dead load (whichever greater)

□ Minimum 15 psf wind

□ Two perpendicular directions

Form Removal

□ Walls/columns: 2× stress rule of thumb

□ Slabs/beams: 70% design strength

□ Field-cured cylinders required

Safety

□ Drawings at jobsite

□ Pre-erection inspection

□ Inspection during placement

□ Reshoring when required

12. Self-Test

1. What is the minimum combined vertical design load for formwork when using motorized carts?

- A) 50 psf
- B) 100 psf
- C) 125 psf
- D) 150 psf

Answer: C) 125 psf

2. For a column form with a plan dimension of 6 ft, which formula applies?

- A) Wall formula (720 constant)
- B) Column formula (785 constant)
- C) Hydrostatic (150h) always
- D) Minimum 600 psf always

Answer: B) Column formula (column = no plan dimension > 6.5 ft) [1]

3. What is the minimum lateral bracing load at a floor edge?

- A) 50 lb/ft
- B) 100 lb/ft or 2% dead load, whichever is greater

- C) 2% dead load only
- D) 15 psf

Answer: B) 100 lb/ft or 2% dead load, whichever is greater

4. When is a written design by a qualified designer required for shoring?

- A) All shoring operations
- B) Tiered single post shores only
- C) Shoring over 16 ft only
- D) Never

Answer: B) Tiered single post shores 30

5. What is the maximum shore load in a two-shore, one-reshore system?

- A) 1.0D
- B) 1.2D
- C) 1.5D
- D) 2.0D

Answer: C) 1.5D

6. At what concrete strength can forms be removed from slabs?

- A) 50% design strength
- B) 70% design strength
- C) 85% design strength
- D) 100% design strength

Answer: B) 70% design strength

7. What is the chemistry coefficient for a lignosulfonate retarder?

- A) 1.0
- B) 1.2
- C) 1.4

- D) 1.5

Answer: A) 1.0

8. A form tie has a tributary area of 6 ft². If Pmax = 1,200 psf, what is the tie tension?

- A) 200 lb
- B) 720 lb
- C) 7,200 lb
- D) 12,000 lb

Answer: C) 7,200 lb (1,200 × 6)

13. Flagged Items – Outside NCEES Scope or Requiring Verification

The following items could not be fully verified against NCEES-published specifications or the PE Civil Reference Handbook, Version 2.0:

1. **Exact USCS column formula constant:** The SI forms clearly show 720 (wall) and 785 (column). The USCS equivalent for columns requires verification against ACI 347R-14 directly. The wall USCS form (150 + 9000R/T) is confirmed .
2. **Specific deflection limits:** ACI 347R provides l/360 for sheathing and l/240 for studs/wales, but the PE Civil Reference Handbook may present these differently. Verify against the handbook during exam preparation.
3. **Reshoring load distribution table values:** The simplified 2/3–1/3 rule is widely used, but the exact ACI 347.2R table should be consulted for detailed analysis. The exam may require direct table lookup.
4. **ASCE 37 load combinations for formwork:** ASCE 37 primarily addresses permanent structures during construction. Its direct application to formwork design is limited; ACI 347R is the primary formwork design reference.
5. **OSHA 1926.703(b)(8) exact wording:** The requirement applies to "single post shores used one on top of another (tiered)." The exam may test this specific condition rather than general shoring.

This review is designed to maximize readiness for the NCEES PE Civil: Construction exam. It does not guarantee a passing score. Always verify formulas and provisions against the current PE Civil Reference Handbook and referenced standards during the exam.

92 web pages

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用户需要一份针对NCEES PE土木考试中"Falsework and Scaffolding"的综合性复习指南，要求紧扣考试大纲和PE Civil Reference Handbook 2.0。这个主题涉及多个技术标准和规范，比如AASHTO、ACI 347、ASCE 37和OSHA 1926。为了高效获取所需信息，我考虑从几个关键维度同时进行搜索，覆盖设计载荷、设计程序、OSHA法规要求、不同标准的差异以及设计示例等核心方面。这样并行搜索可以更快地定位到权威资料，为后续的详细撰写打下基础。

Found 73 web pages

搜索结果回来了，涵盖了几个关键方面。关于设计载荷，有结果提到了AASHTO和ACI 347中关于混凝土单位重量（160 pcf）的要求，以及最小水平荷载为总恒载2%的规定。OSHA 1926 Subpart L的搜索结果明确给出了脚手架组件4:1的安全系数要求。还有关于平台尺寸、护栏高度等具体规定。风荷载方面，有结果提到了重型钢制支撑塔的形状系数2.2和高度分区。抗倾覆稳定性的搜索结果也显示了1.25的安全系数。这些信息比较分散，需要打开几个看起来最权威的页面，比如OSHA的官方eTool和州交通部的规范页面，来获取更详细和准确的条款，以便整合成一份结构化的复习材料。

Read 8 pages

(a) Design: The design of falsework and formwork shall conform to the AASHTO Guide Design Specifications for ...

2. Falsework Design and Construction

eTool : Scaffolding - General Requirements for Scaffolds - Scaffolding eTool 1926.451 - General requirements.

<table><tr><td>Height Zone</td><td>Bent A</td><td>Bent B</td></tr><tr><td></td><td>116719 ft-lb
31ft...

View All

已打开的几个页面提供了关键信息：康涅狄格州和宾夕法尼亚州的规范明确了虚假设计载荷（混凝土单位重量160/130 pcf，最小均布活载20 psf，边缘荷载75 lb/ft，水平荷载2%总恒载）以及工作图纸要求（高度超过14英尺或交通下方需PE签章）。OSHA的eTool和法规条款详细列出了脚手架4:1安全系数、平台尺寸、护栏高度等具体要求。加州交通部的虚假手册示例展示了风荷载计算中形状系数2.2和高度分区压力的应用。这些是核心内容，但还需要补充脚手架高度与底座比例、系杆要求、以及虚假与脚手架在安全系数等方面的具体对比。接下来搜索这些缺口，以确保复习材料覆盖全面。

Found 66 web pages

> 搜索结果覆盖了虚假和脚手架的核心设计参数、OSHA具体要求以及两者在安全系数和用途上的关键区别，信息比较具体。接下来需要将这些分散的条款和公式整合成一份结构清晰的复习指南，重点突出设计荷载、稳定性验算、工作图纸要求以及OSHA脚手架规范中的数字指标，并附上对比表格和练习问题，帮助用户快速掌握考点。

Falsework and Scaffolding for the NCEES PE Civil: Construction Exam

Exam-Focused Review | Design for Support of Construction Loads

Scope Note: This review covers only **Falsework and Scaffolding** within the "Design for Support of Construction Loads" area. Formwork is referenced only where necessary to explain falsework and scaffolding behavior. All content is aligned with the NCEES PE Civil: Construction CBT Exam Specifications and the PE Civil Reference Handbook, Version 2.0.

1. Definitions and Distinctions

1.1 Core Definitions

Term	Definition	Primary Function
Falsework	Any temporary structure used to support the permanent structure until it becomes self-supporting [2]	Support vertical and lateral loads of the permanent structure during construction
Formwork	The total system of support for freshly placed concrete, including sheathing, studs, wales, ties, and bracing	Contain lateral pressure of fresh concrete and shape it
Scaffolding	A temporary structure that functions as a work platform supporting construction personnel, materials, and tools	Provide safe access and working platforms
Shoring	Vertical or inclined support members that carry the weight of formwork and fresh concrete	Transfer loads to ground or lower floors

Term	Definition	Primary Function
Reshoring	Supports installed after formwork removal to distribute loads to multiple floors below	Distribute construction loads through multiple levels

Key distinction: Falsework supports structures (bridges, elevated slabs). Scaffolding supports **people and materials** performing work.

1.2 Primary Functions and Differences

Aspect	Falsework	Scaffolding
Primary purpose	Support permanent structure	Worker access and material storage
Governing code	AASHTO Guide Design Specifications for Bridge Temporary Works; ACI 347	OSHA 1926 Subpart L
Safety factor	Varies by material and component (typically 1.5–2.5 based on allowable stress design)	**4:1** minimum (each component must support 4× maximum intended load) [18]
PE seal requirement	Required when falsework exceeds **14 ft** high or traffic travels under/adjacent [1]	Not typically required for standard scaffolding
Minimum vertical load	100 psf (4.8 kPa) combined dead + live	Per manufacturer rated capacity; must meet 4:1 SF
Wind design	Shape factor 2.2 for heavy-duty steel shoring towers	Per manufacturer or engineering design

1.3 Common Errors and Exam Traps

Error	Correct Approach
Confusing falsework and scaffolding safety factors	Falsework: ASD with material-specific factors; Scaffolding: **4:1** minimum
Assuming PE seal always required	Required only when falsework > 14 ft or traffic under/adjacent
Using scaffolding live load for falsework design	Falsework minimum live load = 20 psf uniform + equipment concentrated + 75 lb/ft at overhangs

2. Falsework Design Loads

2.1 Vertical Dead Loads

Per AASHTO and state DOT specifications, dead loads must include:

- Weight of the falsework itself
- Weight of all construction materials to be supported
- **Concrete unit weight assumptions** (combined concrete, reinforcing steel, prestressing steel, and forms):

Concrete Type	Minimum Assumed Unit Weight
Normal-weight concrete	**160 pcf** (0.16 kip/ft³) [2]
Lightweight concrete	**130 pcf** (0.13 kip/ft³) [2]

High-yield exam trap: The falsework design concrete unit weight is **160 pcf**, not 150 pcf (which is used for formwork lateral pressure calculations). This is one of the most commonly tested distinctions.

2.2 Vertical Live Loads

Live loads consist of:

1. **Actual equipment weight** applied as concentrated loads at points of contact
2. **Uniform load** of not less than **20 psf** applied over the supported area [2]
3. **Edge load** of **75 lb/ft** applied at the outside edge of deck overhangs [2]

2.3 Minimum Combined Vertical Design Load

The minimum combined dead + live vertical design load for falsework is:

$$\boxed{100 \text{ psf (USCS)} \quad \text{or} \quad 4.8 \text{ kPa (SI)}}$$

This minimum applies regardless of the actual calculated load.

2.4 Horizontal Design Loads

The horizontal load for falsework bracing design is the **sum** of:

- Horizontal loads due to equipment
- Construction sequence (including unbalanced hydrostatic forces from fluid concrete)
- Stream flow (when applicable)
- Wind allowance

Minimum requirement: In no case shall the horizontal load to be resisted in any direction be less than **2% of the total dead load** [1] .

Exam note: Some specifications require **3%** for certain conditions (e.g., California Falsework Manual for specific applications). The AASHTO-based minimum is **2%**. Read the problem statement carefully.

2.5 Wind Loads on Heavy-Duty Steel Shoring Towers

For heavy-duty steel shoring towers, wind load design uses:

Parameter	Value
Shape factor	**2.2**

Parameter	Value
Wind impact area	Projected area of tower components
Design wind pressure	From height zone table (varies with height above ground)

Wind pressure height zones (Caltrans example): [48]

Height Zone (ft)	Wind Pressure (psf)	Design Wind Load (psf)
0–30	20	20 × 2.2 = 44
30–50	25	25 × 2.2 = 55
50–100	30	30 × 2.2 = 66
100–105	35	35 × 2.2 = 77

Critical exam point: The **shape factor of 2.2** is specifically for heavy-duty steel shoring towers. Conventional falsework uses a drag coefficient $Q = 1 + 0.2W$ (where W is tower width), capped at 10 [49] .

2.6 Common Errors and Exam Traps – Falsework Loads

Error	Correct Approach
Using **150 pcf** instead of **160 pcf**	Falsework concrete unit weight = 160 pcf (normal-weight)
Forgetting **minimum combined load of 100 psf**	Apply 100 psf regardless of calculated load
Using **2% vs. 3%** for horizontal bracing	AASHTO minimum = 2% of total dead load
Forgetting **edge load of 75 lb/ft**	Required at outside edge of deck overhangs
Neglecting **wind on heavy-duty towers**	Shape factor = 2.2; use height zone pressure table
Using scaffolding live load for falsework	Falsework uniform live load minimum = 20 psf

3. Falsework Design Procedure

3.1 Step-by-Step Procedure

Step 1 – Identify governing codes and standards

- AASHTO Guide Design Specifications for Bridge Temporary Works
- ACI 347 (formwork)
- State DOT specifications (if applicable)

Step 2 – Determine dead loads

- Falsework self-weight
- Concrete: 160 pcf (normal-weight) or 130 pcf (lightweight)
- Reinforcing and prestressing steel
- Forms

Step 3 – Determine live loads

- Equipment as concentrated loads
- Uniform load ≥ 20 psf
- Edge load = 75 lb/ft at overhangs

Step 4 – Check minimum combined load

- Verify total ≥ 100 psf

Step 5 – Determine horizontal/bracing loads

- Sum of equipment, construction sequence, stream flow, and wind
- Minimum **2% of total dead load** in any direction

Step 6 – Check wind loads

- Heavy-duty steel shoring: shape factor 2.2
- Use height zone wind pressure table

- Calculate overturning moment

Step 7 – Design structural components

- Check bending, shear, deflection, and buckling
- Use ASD with appropriate allowable stresses

Step 8 – Check overturning stability

- Factor of safety (FS) = **1.25** minimum [58]
- Cable bracing designed to resist 1.25 × total overturning moment minus tower resistance

Step 9 – Prepare working drawings

- Required when falsework exceeds **14 ft** high or traffic travels under/adjacent
- Must be signed and sealed by a PE [2]

3.2 Working Drawings Requirement

Condition	Working Drawings Required?	PE Seal Required?
Falsework ≤ 14 ft, no traffic below	No (unless specified)	No
Falsework > 14 ft	**Yes**	**Yes**
Traffic travels under or adjacent	**Yes**	**Yes**
Both conditions	**Yes**	**Yes**

Working drawings must include:

- Sequence, method, and rate of concrete placement
- Manufacturer catalog cuts for manufactured parts
- Anticipated settlements and deflections
- Falsework footing details [1]

3.3 Common Errors and Exam Traps – Design Procedure

Error	Correct Approach
Forgetting PE seal requirement at 14 ft	Working drawings signed/sealed by PE required when > 14 ft
Missing overturning FS = 1.25	Cable bracing designed for 1.25 × overturning moment
Neglecting construction sequence loads	Include unbalanced hydrostatic forces from fluid concrete
Skipping minimum combined load check	100 psf minimum always applies

4. Scaffolding Design Requirements (OSHA 1926 Subpart L)

4.1 Capacity Requirements

Fundamental OSHA requirement:

$$\boxed{\text{Each scaffold component must support } 4\times \text{ the maximum intended load}}$$

Per 29 CFR 1926.451(a)(1): "Each scaffold and scaffold component shall be capable of supporting, without failure, its own weight and at least 4 times the maximum intended load applied or transmitted to it." [19]

Exceptions:

- Suspension ropes: 6:1 safety factor [19]
- Stall load of hoist: shall not exceed 3× rated load [19]

4.2 Platform Requirements

Requirement	Specification
Maximum distance to structure	**14 inches** (unless guardrails or PFAS used)

Requirement	Specification
Minimum extension over support	**6 inches**
Maximum extension over support	**12 inches** (for platforms ≤ 10 ft); 18 inches for platforms > 10 ft (if not cleated)
Maximum gap between platform and uprights	1 inch (9.5 inches with side brackets) [18]
Minimum platform width	**18 inches** [18]
Maximum deflection	1/60 of span [18]
Guardrail height	**38–45 inches** above platform surface
Mid-rail	Approximately midway between top rail and platform
Toeboard height	**4 inches** minimum (typically)
Safe access	Within **2 ft** of platform
Toeboard	Required when platform > 10 ft above lower level

4.3 Access Requirements

Requirement	Specification
Climbing braces	Prohibited as means of access
Stair systems	Required above **6 ft**
Ladders	Must be positioned to prevent slipping
Access within	**2 ft** of platform

4.4 Stability Requirements

Requirement	Specification
Height-to-base ratio	Ties required when > **4:1**
Top tie	Placed no further than **4:1** ratio from top
Vertical tie spacing (tube & coupler)	**3 ft** for narrow bases (≤ 3 ft)
Bracing	Two mutually perpendicular directions

4.5 Training Requirements

Per **29 CFR 1926.454**:

- Employer must have each employee trained by a **qualified person**
- Training must cover hazards associated with the type of scaffold
- Topics include:
  - Electrical hazards
  - Fall hazards
  - Falling object hazards
  - Correct erection/dismantling procedures
  - Maximum intended load and rated capacities
  - Proper use of scaffolds

4.6 Common Errors and Exam Traps – Scaffolding

Error	Correct Approach
Using 2:1 or 3:1 safety factor	**4:1** for scaffolds (6:1 for suspension ropes)
Forgetting 14-inch maximum distance to structure	Platform front edge ≤ 14 inches from work
Confusing 6-inch minimum and 12-inch maximum extension	End must extend **6–12 inches** over support

Error	Correct Approach
Missing tie-in requirement at 4:1	Height-to-base ratio > 4:1 requires ties
Using climbing braces for access	**Prohibited**; stair systems required above 6 ft
Forgetting guardrail height range	**38–45 inches** (not 42 inches fixed)

5. Falsework vs. Scaffolding: Key Differences Summary

Parameter	Falsework	Scaffolding
Primary purpose	Support structure	Worker/materials platform
Governing code	AASHTO Guide; ACI 347	OSHA 1926 Subpart L
Safety factor	Per ASD (material-specific)	**4:1** minimum
Minimum vertical load	**100 psf** combined	Per rated capacity; must meet 4:1 SF
Live loads	≥ 20 psf uniform + equipment concentrated + 75 lb/ft at overhangs	Per manufacturer; typically 25–75 psf
PE seal required	Yes, if > **14 ft** or traffic below	Not typically
Wind design	Shape factor **2.2** (heavy-duty steel towers)	Per manufacturer
Horizontal design	**2%** of total dead load minimum	Tie-in requirements at 4:1 ratio
Concrete unit weight	**160 pcf** (normal-weight)	N/A

Common Errors and Exam Traps – Comparisons

Error	Correct Approach
Confusing 160 pcf (falsework) with 150 pcf (formwork)	Falsework: 160 pcf; Formwork pressure: 150 pcf hydrostatic
Applying 4:1 SF to falsework	4:1 is **scaffolding**; falsework uses ASD
Applying PE seal to all falsework	Only when > 14 ft or traffic below
Using scaffolding tie-in rules for falsework	Different stability requirements

6. Worked Examples and Practice Problems

Problem 1 – Falsework Horizontal Bracing Load

Problem: A falsework system supports a concrete bridge deck. The total dead load (concrete + falsework + forms) = 200,000 lb. What is the minimum horizontal design load for the bracing system?

Solution:

1. Minimum horizontal load = **2% of total dead load** [1]
2. $H_{\min} = 0.02 \times 200,000 = 4,000$ lb
3. **Answer: 4,000 lb** in any direction

Problem 2 – Scaffolding Platform Capacity

Problem: A scaffold platform is designed to carry a maximum intended load of 500 lb per platform section. What is the minimum capacity each component must support?

Solution:

1. OSHA requires each component to support **4× the maximum intended load** [18]
2. $\text{Required capacity} = 4 \times 500 = 2,000$ lb
3. **Answer: 2,000 lb**

Problem 3 – Wind Load on Heavy-Duty Shoring Tower

Problem: A heavy-duty steel shoring tower is 80 ft tall. The projected area of the tower face is 2.0 ft²/ft of height. The wind pressure from the height zone table is 25 psf for the 50–100 ft zone. What is the design wind load per foot of tower height?

Solution:

1. Shape factor for heavy-duty steel shoring = **2.2**
2. Design wind pressure = $25 \times 2.2 = 55$ psf
3. Wind load per foot = $55 \text{ psf} \times 2.0 \text{ ft}^2/\text{ft} = 110$ lb/ft
4. **Answer: 110 lb/ft of tower height**

Problem 4 – Overturning Stability Check

Problem: A falsework tower has an overturning moment of 100,000 ft-lb due to wind. The resisting moment from the tower's dead load is 80,000 ft-lb. Is the tower stable without cable bracing? If not, what is the required cable resistance?

Solution:

1. Required FS against overturning = **1.25** [58]
2. Required resisting moment = $1.25 \times 100,000 = 125,000$ ft-lb
3. Available resisting moment = 80,000 ft-lb < 125,000 ft-lb → **Not stable**
4. Required cable resistance = $125,000 - 80,000 = 45,000$ ft-lb
5. **Answer: Tower requires cable bracing capable of resisting 45,000 ft-lb**

Problem 5 – PE Seal Requirement

Problem: A contractor proposes falsework that is 12 ft high with no traffic traveling under or adjacent to it. Is a PE-sealed working drawing required?

Solution:

1. Working drawings with PE seal required when falsework exceeds **14 ft** or traffic travels under/adjacent [2]

2. 12 ft < 14 ft and no traffic → **PE seal NOT required**
3. **Answer: No PE-sealed working drawings required** (unless otherwise specified by contract)

Problem 6 – Scaffolding Platform Extension

Problem: A scaffold platform is 8 ft long. The plank is not cleated or hooked. How much must it extend over the support?

Solution:

1. Minimum extension = **6 inches** over centerline of support
2. Maximum extension for platform ≤ 10 ft = **12 inches**
3. **Answer: Between 6 and 12 inches**

7. Safety and Code Conformance

7.1 OSHA 1926 Subpart L – Scaffolding (Key Provisions)

Section	Requirement
1926.451(a)(1)	4:1 capacity for all components [19]
1926.451(a)(6)	Designed by qualified person [19]
1926.451(b)(1)	Fully planked/decked platforms
1926.451(b)(4)	6-inch minimum extension
1926.451(b)(5)	12-inch maximum extension (≤ 10 ft platforms)
1926.451(c)(1)	Tie-in required at > 4:1 height-to-base ratio
1926.454	Training requirements

7.2 OSHA 1926 Subpart Q – Concrete and Masonry Construction

- Formwork must support all anticipated vertical and lateral loads
- Drawings/plans available at jobsite
- Shoring equipment inspected before, during, and after concrete placement
- Tiered single post shores: written design by qualified designer

7.3 AASHTO Guide Design Specifications for Bridge Temporary Works

- Primary design reference for falsework
- Load factors per AASHTO LRFD Bridge Design Specifications
- All applicable load combinations must be investigated [1]

7.4 ACI 347 – Guide to Formwork for Concrete

- Lateral pressure formulas for wall and column forms
- Minimum design loads (100 psf, 125 psf for buggies)
- Deflection limits for formwork components

7.5 ASCE 37 – Design Loads on Structures During Construction

- Specifies construction loads for temporary structures
- Provides load combinations for construction-stage analysis
- Referenced by NCEES as applicable standard

7.6 Common Errors – Safety and Code

Error	Correct Approach
Missing PE seal for falsework > 14 ft	Required by AASHTO and state DOTs
Using 150 pcf instead of 160 pcf	Falsework concrete unit weight = **160 pcf**
Forgetting 2% horizontal load minimum	Minimum horizontal bracing load = 2% of total dead load

Error	Correct Approach
Neglecting 1.25 overturning FS	Cable bracing designed for 1.25 × overturning moment
Missing OSHA 4:1 scaffolding SF	Required for all scaffold components

8. Common Errors and Exam Traps – Master List

#	Trap	Correct Approach
1	Unit inconsistencies (USCS vs. SI)	Check units in every formula
2	Using **150 pcf** instead of **160 pcf**	Falsework concrete = 160 pcf (normal-weight)
3	Forgetting minimum combined load of **100 psf**	Always apply 100 psf minimum
4	Using **2% vs. 3%** for horizontal bracing	AASHTO minimum = **2%** of total dead load
5	Forgetting **edge load of 75 lb/ft**	Required at outside edge of deck overhangs
6	Confusing **falsework and scaffolding safety factors**	Falsework: ASD; Scaffolding: **4:1**
7	Forgetting PE seal requirement at **14 ft**	Required when falsework > 14 ft or traffic below
8	Neglecting **wind on heavy-duty towers**	Shape factor = **2.2**
9	Forgetting **overturning FS = 1.25**	Cable bracing designed for 1.25 × moment

#	Trap	Correct Approach
10	Misidentifying when working drawings required	> 14 ft OR traffic under/adjacent
11	Misunderstanding OSHA platform requirements	14-in max to structure; 6-in min extension; 12-in max extension
12	Using **2:1 or 3:1** safety factor for scaffolding	**4:1** required
13	Forgetting **tie-in at 4:1 height-to-base ratio**	Required for supported scaffolds > 4:1
14	Using climbing braces for access	**Prohibited**; stairs required above 6 ft
15	Forgetting **20 psf minimum live load** for falsework	Required by AASHTO

9. High-Yield Checklist

Falsework Loads

□ Concrete unit weight = **160 pcf** (normal-weight) or 130 pcf (lightweight)

□ Minimum combined vertical load = **100 psf**

□ Uniform live load ≥ **20 psf**

□ Edge load = **75 lb/ft** at overhangs

□ Horizontal load ≥ **2% of total dead load**

□ Wind shape factor = **2.2** for heavy-duty steel shoring

Falsework Design

□ Working drawings required when > **14 ft** or traffic under/adjacent

□ PE seal required for working drawings

□ Overturning FS = **1.25**

□ Cable bracing designed for 1.25 × overturning moment

Scaffolding (OSHA 1926 Subpart L)

□ Safety factor = **4:1** for all components

□ Platform distance to structure ≤ **14 inches**

□ Platform extension: **6–12 inches**

□ Platform width ≥ **18 inches**

□ Guardrail height: **38–45 inches**

□ Tie-in required at **4:1** height-to-base ratio

□ Tube & coupler vertical tie spacing = **3 ft** (narrow bases)

□ Access within **2 ft** of platform

□ Climbing braces prohibited

□ Training required per **1926.454**

Comparisons

□ Falsework vs. formwork concrete unit weight (160 vs. 150 pcf)

□ Falsework vs. scaffolding safety factors (ASD vs. 4:1)

□ PE seal only for falsework > 14 ft

10. Self-Test

1. What is the minimum combined vertical design load for falsework?

- A) 50 psf
- B) 75 psf
- C) 100 psf
- D) 125 psf

Answer: C) 100 psf

2. What concrete unit weight must be assumed for falsework design with normal-weight concrete?

- A) 145 pcf
- B) 150 pcf
- C) 160 pcf
- D) 165 pcf

Answer: C) 160 pcf

3. What is the minimum safety factor for scaffolding components per OSHA?

- A) 2:1
- B) 3:1
- C) 4:1
- D) 6:1

Answer: C) 4:1

4. When are PE-sealed working drawings required for falsework?

- A) Always
- B) When falsework exceeds 10 ft
- C) When falsework exceeds 14 ft or traffic travels under/adjacent
- D) Never

Answer: C) When falsework exceeds 14 ft or traffic travels under/adjacent

5. What is the shape factor for heavy-duty steel shoring towers?

- A) 1.0
- B) 1.5
- C) 2.0
- D) 2.2

Answer: D) 2.2

6. What is the minimum horizontal bracing load for falsework?

- A) 1% of total dead load
- B) 2% of total dead load
- C) 3% of total dead load
- D) 5% of total dead load

Answer: B) 2% of total dead load

7. What is the maximum distance a scaffold platform may be from the structure?

- A) 6 inches
- B) 12 inches
- C) 14 inches
- D) 18 inches

Answer: C) 14 inches

8. What is the required overturning factor of safety for falsework cable bracing?

- A) 1.0
- B) 1.25
- C) 1.5
- D) 2.0

Answer: B) 1.25

9. At what height-to-base ratio are tie-ins required for supported scaffolds?

- A) 2:1
- B) 3:1
- C) 4:1
- D) 6:1

Answer: C) 4:1

10. What is the minimum extension of a scaffold platform over its support?

- A) 3 inches
- B) 6 inches
- C) 12 inches
- D) 18 inches

Answer: B) 6 inches

11. Flagged Items – Outside NCEES Scope or Requiring Verification

The following items could not be fully verified against NCEES-published specifications or the PE Civil Reference Handbook, Version 2.0:

1. **Exact falsework safety factors by material:** AASHTO and state DOTs use ASD with material-specific factors (typically 1.5–2.5 for wood, 1.67 for steel). The exam may test conceptual understanding rather than specific numeric factors.
2. **SI unit equivalents:** The SI minimum combined load is 4.8 kPa (≈100 psf). Other SI conversions should be verified against the PE Civil Reference Handbook.
3. **State-specific horizontal load percentages:** While AASHTO minimum is 2%, some states (e.g., California) may require 3% for certain conditions. The exam should specify which standard governs.
4. **Tube and coupler scaffold tie spacing:** The 3-ft vertical tie spacing for narrow bases (≤ 3 ft) is from OSHA non-mandatory Appendix E. Verify applicability to specific exam problems .
5. **Scaffolding toeboard height:** OSHA does not specify an exact toeboard height in all cases; typical practice is 4 inches minimum. Verify against 1926.451(h).
6. **ASCE 37 load combinations for falsework:** ASCE 37 primarily addresses permanent structures during construction. AASHTO and ACI 347 are the primary references for falsework design loads.

This review is designed to maximize readiness for the NCEES PE Civil: Construction exam. It does not guarantee a passing score. Always verify formulas and provisions against the current PE Civil Reference Handbook and referenced standards during the exam.

132 web pages

Act as a licensed PE an...
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用户需要一份关于Shoring and Reshoring的考试复习资料，要求非常具体，必须基于ACI 347R、OSHA 1926 Subpart Q等标准。这份资料需要覆盖定义、荷载、设计、安全等多个方面，并且要提供公式和例题。为了高效地收集所需信息，我将同时进行多项搜索，以覆盖这些关键领域。

Found 73 web pages

所有并行搜索已完成。现在需要整合这些结果，形成一份全面的复习资料。为了获取更详细的信息，我需要同时打开几个最有用的链接。

Read 5 pages

ACI PRC-347.2-17(25): Guide for Shoring/Reshoring of Concrete Multistory Buildings (Reapproved 2025)

1926.703 - Requirements for cast-in-place concrete. - - Part Number:1926

Table 5.5 — Simplified Analysis of Loads on Shores and Slabs of Multistory Structure

29 CFR § 1926.703 - Requirements for cast-in-place concrete. - 29 CFR § 1926.703 - Requirements for cast-in-pl...

The maximum shoring and reshoring construction load occurs during the placement of the top floor level

这些资料提供了关于ACI 347.2R、OSHA要求和荷载分布表的详细信息。为了全面覆盖用户要求的各个子主题，我们需要同时进行多项搜索，以获取关于定义、设计、安全、施工应用等方面的具体信息。

Found 76 web pages

所有并行搜索已完成。现在需要整合这些信息，形成一份全面的复习资料。我将组织内容，涵盖定义、荷载、设计、安全、施工应用等各个方面，并包含公式、例题和考试技巧。

Shoring and Reshoring for the NCEES PE Civil: Construction Exam

Exam-Focused Review | Design for Support of Construction Loads

Scope Note: This review covers only **Shoring and Reshoring** within the "Design for Support of Construction Loads" area. Formwork, falsework, and scaffolding are referenced only where necessary to explain shoring and reshoring behavior. All content is aligned with the NCEES PE Civil: Construction CBT Exam Specifications and the PE Civil Reference Handbook, Version 2.0.

1. Definitions and Distinctions

1.1 Core Definitions

Term	Definition	Primary Function
Shoring	Vertical or inclined support members designed to carry the weight of formwork, fresh concrete, and construction loads above	Transfer vertical loads from formwork/falsework to ground or lower floors
Reshoring	Construction operation in which shoring equipment (reshores) is placed as the original forms and shores are removed, to support partially cured concrete and construction loads	Distribute construction loads across multiple floors
Backshoring (Preshoring)	Shores placed snugly under a concrete slab after original formwork and shores have been removed from a small area at a time, without allowing the slab to deflect	Maintain continuous support during form removal sequence
Formwork	Total system of support for freshly placed concrete, including sheathing, studs, wales, ties, and bracing	Contain lateral pressure and shape concrete

Term	Definition	Primary Function
Falsework	Any temporary structure used to support the forms for concrete	Support formwork and fresh concrete

Key distinction: Shoring supports vertical loads during concrete placement. Reshoring is **installed after** formwork removal to distribute loads to multiple floors below. Backshoring is installed **sequentially** as forms are removed, maintaining continuous contact.

1.2 Differences Between Shoring, Reshoring, Formwork, Falsework, and Scaffolding

Aspect	Shoring	Reshoring	Formwork	Falsework	Scaffolding
Primary purpose	Support vertical loads	Distribute loads across floors	Contain and shape concrete	Support forms	Worker access/platform
Timing	During concrete placement	After form removal	During concrete placement	During construction	During construction
Governing code	ACI 347.2R, OSHA 1926.703	ACI 347.2R, OSHA 1926.703	ACI 347R, OSHA 1926.703	AASHTO, OSHA 1926.703	OSHA 1926 Subpart L
Safety factor	ASD (material-specific)	ASD (material-specific)	ASD (material-specific)	ASD (material-specific)	4:1 minimum
PE seal	Required for tiered shores	Required for tiered shores	Per formwork requirements	Required >14 ft or traffic below	Not typically

1.3 Common Errors and Exam Traps

Error	Correct Approach
Confusing shoring and reshoring definitions	Shoring: during placement; Reshoring: after form removal

Error	Correct Approach
Treating backshoring and reshoring as identical	Backshoring: sequential, no deflection allowed; Reshoring: after full removal
Assuming shoring and falsework are the same	Shoring = vertical supports; Falsework = entire temporary support structure

2. Load Path and Multistory Construction

2.1 Load Path from Fresh Concrete to Lower Floors

The construction load path follows this sequence:

1. **Fresh concrete + formwork** → supported by **shores**
2. **Shores** → transfer load to **floor below** (slab)
3. **Floor below** → deflects and distributes load to **reshores below**
4. **Reshores** → transfer load to **next floor down**
5. Process continues through multiple floors until load reaches **ground**

The key principle: **Freshly placed concrete cannot support its own weight.** All construction loads must be carried by shores and reshores until concrete gains sufficient strength.

2.2 Simplified Load Distribution for Shoring and Reshoring

ACI 347.2R provides a simplified method for load distribution based on the Grundy and Kabaila (1963) approach. The key findings are:

Maximum shore/reshore construction load = 1.5D (or 169 psf) for both three-reshore and two-reshore systems [66] .

Reshore load distribution:

- **Top reshore** carries **2/3** of the load
- **Lower reshore/slab** carries **1/3** of the load

100% reshoring plan: Install **8 reshores for every 10 shores removed** to maintain adequate load distribution .

2.3 Table 5.5 – Simplified Analysis of Loads (ACI 347.2R)

The following table illustrates how loads develop in a two-shore, one-reshore system for a flat-slab building (D = weight of one slab) [2] :

Step	Operation	Fresh Slab Load	Slab 1 Load	Slab 2 Load	Shore Load
1	Place Level 1 concrete	1D	0	0	1D
2	Place Level 2 concrete	1D	0	0	2D
3	Remove first-level shores	1D	1D	1D	0
4	Place reshores (snug)	1D	1D	1D	0
5	Place Level 3 concrete	1D	1D	1D	1D
6	Remove first-floor reshores	1.33D	1.33D	1.33D	0.67D

High-yield exam point: The maximum load on any shore or reshore is **1.5D**, occurring during the placement of the top floor level [66] .

2.4 Effect of Reshoring on Slab Loading and Curing

- **Reshoring allows slabs to work together** to support construction loads, reducing the maximum load on any single slab
- Without reshoring, the freshly cast slab must carry its own weight plus construction loads alone
- Reshoring enables **earlier form removal** because the concrete only needs to reach **70% of design strength** (not full strength)
- **Slab capacity check:** The load on each slab must not exceed its available capacity at the time of loading, accounting for concrete maturity and temperature

2.5 Number of Floors Required to Support Construction Loads

The number of floors required depends on:

- **Construction cycle time** (7-day, 10-day, 15-day cycles)
- **Concrete strength gain** (affected by temperature)
- **Shoring/reshoring scheme** (two-reshore vs. three-reshore system)
- **Construction live load** (50 psf or 100 psf)

Construction Cycle	Design Live Load	Typical Reshore Levels Required
7-day cycle	50 psf	2–3 levels
7-day cycle	100 psf	3 levels
10-day cycle	50 psf	2 levels
10-day cycle	100 psf	2–3 levels
15-day cycle	50 psf	2 levels
15-day cycle	100 psf	2 levels

2.6 Common Errors and Exam Traps – Load Path

Error	Correct Approach
Assuming fresh concrete carries its own weight	Fresh concrete cannot support itself; shores carry all load
Forgetting the 1.5D maximum shore load	Maximum shore/reshore load = 1.5D
Using 1/2–1/2 reshore distribution	Top reshore = 2/3; lower = 1/3
Overlooking slab capacity checks	Each slab must be checked against construction loads
Misunderstanding 100% reshoring plan	8 reshores per 10 shores removed

Error	Correct Approach
Forgetting temperature effects on strength gain	Cold weather slows curing → longer reshore duration

3. Design Requirements and Loads

3.1 Vertical Loads

Load Component	Value / Notes
Concrete unit weight	150 pcf (normal-weight); 145 pcf for formwork pressure
Reinforcement	Included in concrete unit weight for construction loads
Formwork self-weight	Typically 6.5 psf for slab formwork
Falsework/shoring self-weight	Varies by system; must be included
Live load – hand tools	Minimum **50 psf**
Live load – motorized carts	Minimum **75 psf**
Minimum combined vertical design load	**100 psf** for hand tools; **125 psf** for motorized carts

Critical exam point: The minimum combined vertical design load of **100 psf** applies regardless of slab thickness or calculated load. For motorized carts/buggies, the minimum is **125 psf**.

3.2 Lateral Loads

Load Type	Requirement
Wind	Minimum **15 psf** on formwork/shoring exposed to wind

Load Type	Requirement
Construction sequence	Include unbalanced loads from concrete placement
Equipment	Include lateral forces from equipment operation
Minimum bracing	**100 lb/ft** along slab edge or **2% of supported vertical load**, whichever is greater
Wall form bracing	Minimum **100 lb/ft** of wall

Exam trap: The 2% horizontal load requirement is a **minimum**. Some specifications require **3%** for certain conditions. Read the problem statement carefully.

3.3 Design of Shoring Members

Shoring members (posts, columns) are designed for **axial compression** and must be checked for:

Axial capacity:

$$P_{\text{allow}} = F'_c \times A$$

Where F'_c = adjusted allowable compressive stress

Buckling (Euler equation for long columns):

$$P_{\text{cr}} = \frac{\pi^2 E I}{(K L)^2}$$

Combined stresses (NDS interaction):

$$\left(\frac{f_c}{F'_c}\right)^2 + \frac{f_b}{F'_b} \leq 1.0$$

For wood shoring, use NDS adjustment factors:

- **CD = 1.25** for construction loads lasting ≤ 7 days
- **CM** = wet service factor (1.0 for dry conditions)
- **Cp** = column stability factor

For steel shoring: Use AISC ASD with appropriate allowable stresses.

3.4 Design of Reshoring Systems

Reshoring design requires:

1. **Determine maximum shore/reshore load** = 1.5D
2. **Check slab capacity** at each level for the construction load
3. **Verify reshore capacity** ≥ 2/3 of total load (top reshore)
4. **Check bearing** at reshore heads
5. **Ensure vertical alignment** of reshores and shores

3.5 Worked Example – Shoring Design

Problem: A 10-in. thick slab is supported by wood shores. Slab weight = 125 psf. Construction live load = 50 psf. Shore spacing = 4 ft × 4 ft. Wood is Douglas Fir-Larch (Fc = 1,500 psi, E = 1,700,000 psi). Shore is 10 ft tall, 4×4 (A = 12.25 in², I = 12.5 in⁴). Check axial capacity.

Solution:

1. **Tributary area** = 4 × 4 = 16 ft²
2. **Load per shore** = (125 + 50) × 16 = 2,800 lb
3. **Slenderness ratio** (K = 1.0 for pinned-pinned): $KL/d = 120/3.5 = 34.3$ (intermediate column)
4. **Euler critical stress:** $F_e = \frac{\pi^2 E}{(KL/d)^2} = \frac{\pi^2 \times 1{,}700{,}000}{(34.3)^2} = 14,260$ psi
5. **Column stability factor:** $C_p = 0.8$ (approximate for intermediate column)
6. **Adjusted allowable stress:** $F_c' = 1,500 \times 1.25 \times 0.8 = 1,500$ psi
7. **Allowable load:** $P = 1,500 \times 12.25 = 18,375$ lb
8. **Check:** 18,375 lb > 2,800 lb ✓ **Adequate**

3.6 Common Errors and Exam Traps – Design

Error	Correct Approach
Forgetting minimum combined load	100 psf minimum (125 psf for carts)
Neglecting buckling for long shores	Check slenderness; use NDS column equations
Using CD = 1.0 for construction loads	CD = 1.25 for ≤ 7-day construction loads
Forgetting 2% horizontal load	Minimum bracing = 2% of supported vertical load
Overlooking slab capacity for reshores	Check each slab against construction loads

4. OSHA Requirements for Shoring and Reshoring

4.1 OSHA 1926 Subpart Q – Key Provisions

Section	Requirement
1926.703(a)(1)	Formwork must support all anticipated vertical and lateral loads without failure [16]
1926.703(a)(2)	Drawings/plans for formwork and shoring must be available at jobsite [16]
1926.703(b)(1)	All shoring equipment inspected before erection [16]
1926.703(b)(2)	Damaged shoring equipment shall not be used [16]
1926.703(b)(3)	Erected shoring inspected before, during, and after concrete placement [16]
1926.703(b)(5)	Sills must be sound, rigid, and capable of carrying maximum intended load [16]
1926.703(b)(6)	Base plates, shore heads, and adjustment screws in firm contact [16]
1926.703(b)(7)	Eccentric loads prohibited unless designed for [16]

Section	Requirement
1926.703(b)(8)	**Tiered single post shores** – special requirements [16]
1926.703(b)(10)	Reshoring erected as original forms removed when concrete must support loads exceeding capacity [16]

4.2 Written Design Requirement for Tiered Shores

OSHA 1926.703(b)(8)(i): Whenever single post shores are used one on top of another (tiered), the design of the shoring shall be prepared by a **qualified designer** and the erected shoring shall be inspected by an **engineer qualified in structural design** [16] .

Additional requirements for tiered shores:

Requirement	Specification
Vertical alignment	Single post shores shall be vertically aligned [16]
Splicing	Shores shall be spliced to prevent misalignment [16]
Bracing	Adequately braced in two mutually perpendicular directions at splice level; each tier diagonally braced [16]
Adjustment	Adjustment to raise formwork shall not be made after concrete placement [16]

High-yield exam trap: The written design requirement applies to **tiered single post shores** – not all shoring. However, drawings/plans for all formwork and shoring must be available at the jobsite at all times.

4.3 Inspection and Maintenance Requirements

- **Pre-erection inspection:** All shoring equipment must be inspected before erection
- **During placement:** Inspection immediately prior to, during, and after concrete placement
- **Post-erection:** Damaged or weakened shoring must be immediately reinforced
- **Damaged equipment:** Shoring with reduced strength shall not be used

4.4 Common Errors – OSHA Requirements

Error	Correct Approach
Missing written design for tiered shores	Required by OSHA 1926.703(b)(8)(i)
Skipping pre-erection inspection	Required for all shoring equipment
No inspection during placement	Required immediately prior to and during
Using damaged shoring equipment	Prohibited if strength is reduced
Forgetting drawings at jobsite	Required by OSHA 1926.703(a)(2)

5. Form Removal and Reshoring Criteria

5.1 Minimum Time and Strength Requirements

Member Type	Minimum Strength for Form Removal
Walls, columns	Concrete must be adequately set; 2× stress at removal (rule of thumb)
Slabs, beams (vertical supports)	**70% of specified 28-day compressive strength**
Structural members	Must support own weight + construction loads

Rule of thumb: Concrete strength at form removal ≥ **2 ×** the stress to which it will be subjected at removal.

ACI 347R: Supporting forms and shores must not be removed from horizontal members until concrete strength is at least **70% of design strength**, as determined by field-cured cylinders .

5.2 Member-Specific Removal Criteria

Member	Typical Removal Time	Strength Requirement
Walls	12–24 hours	2× stress at removal
Columns	24–48 hours	2× stress at removal
Slab soffits (props left)	3–4 days	70% design strength
Beam soffits (props left)	7 days	70% design strength
Props under slabs	14 days	70% design strength
Props under beams	21 days	70% design strength

5.3 Reshoring Placement and Timing

- **Reshores must be erected as original forms and shores are removed** whenever concrete is required to support loads exceeding its capacity [16]
- Reshores must be placed **snugly** under the slab – not initially loaded
- **100% reshoring plan:** 8 reshores per 10 shores removed
- Reshores shall remain until concrete has attained the **specified 28-day compressive strength**

5.4 Common Errors – Form Removal

Error	Correct Approach
Removing forms without cylinder tests	Field-cured cylinders required for horizontal members
Confusing 2× rule with 70% rule	2× stress = walls/columns; 70% design strength = slabs/beams
Not reshoring when required	Reshore whenever concrete must support excess loads
Removing props under beams too early	21 days typical for beams
Assuming reshores are initially loaded	Reshores placed snug but not loaded

6. Construction Applications and Quality Control

6.1 Shoring and Reshoring for Multistory Buildings

- **Construction cycle time** affects the number of floors required
- **Temperature** affects concrete strength gain and reshore duration
- **Post-tensioned construction** requires special consideration for load redistribution
- **Slab capacity** must be checked at each level during construction

6.2 Shoring for Excavation Support and Earth Retention

- OSHA 1926 Subpart P governs excavation shoring
- Timber shoring, aluminum hydraulic shoring, and other systems are covered
- Protective systems required for excavations **5 ft or deeper**
- Excavations **20 ft or deeper** require design by a registered PE

6.3 Inspection, Tolerances, and Specification Conformance

- **Pre-erection inspection** of all shoring components
- **During placement inspection** – immediately prior to, during, and after
- **Tolerances:** Vertical alignment, plumbness, and elevation must be within specification
- **Documentation:** Drawings and plans available at jobsite

6.4 Documentation and Quality Control

- Shoring must be installed per authorized shop drawings
- Components inspected prior to erection; heavily rusted, bent, or defective components rejected
- **Independent design check** and **independent QC** may be required
- As-built inspection by engineer in responsible charge

6.5 Common Field and Design Errors

Error	Consequence
Inadequate reshore distribution	Slab overloading
Missing pre-erection inspection	Use of damaged equipment
No slab capacity check	Potential slab failure
Improper vertical alignment	Eccentric loading and buckling
Insufficient bracing	Lateral instability

7. Safety and Code Conformance

7.1 OSHA 1926 Subpart Q – Concrete and Masonry Construction

Key provisions for shoring and reshoring are detailed in Section 4 above.

7.2 OSHA 1926 Subpart P – Excavations

Requirement	Specification
Protective systems	Required for excavations 5 ft or deeper
Design by PE	Required for excavations 20 ft or deeper
Timber shoring tables	Provided in Subpart P Appendix C
Aluminum hydraulic shoring	Per Appendix D

7.3 ACI 347.2R – Guide for Shoring/Reshoring of Concrete Multistory Buildings

- Primary reference for shoring/reshoring of multistory buildings [1]

- Provides simplified method for load distribution
- Includes construction load factors and early-age strength requirements
- Design examples provided

7.4 ASCE 37 – Design Loads on Structures During Construction

- Defines minimum design loads during construction
- Construction live loads vary from **20 psf to 70 psf**
- Covers dead loads, construction loads, lateral earth pressure, and environmental loads

7.5 ACI 318 and AISC Provisions

- **ACI 318:** Concrete member behavior, reinforcement properties, and design provisions relevant to shoring
- **AISC:** Steel shoring member design, fastener groups, and welding symbols

7.6 Common Errors – Safety and Code

Error	Correct Approach
Missing PE seal for tiered shores	Required by OSHA 1926.703(b)(8)(i)
Forgetting OSHA Subpart P for excavation shoring	Protective systems required at 5 ft depth
Using ASCE 37 loads without checking ACI 347.2R	ACI 347.2R is primary for multistory shoring
Neglecting construction load factors	Include impact and dynamic effects

8. Common Errors and Exam Traps – Master List

#	Trap	Correct Approach
1	Unit inconsistencies (USCS vs. SI)	Check units in every formula

#	Trap	Correct Approach
2	Confusing shoring and reshoring definitions	Shoring: during placement; Reshoring: after removal
3	Forgetting written design for tiered shores	Required by OSHA 1926.703(b)(8)(i)
4	Misunderstanding load distribution	Maximum shore load = 1.5D; top reshore = 2/3
5	Overlooking slab capacity checks	Each slab must support construction loads
6	Neglecting minimum combined vertical load	100 psf (125 psf for motorized carts)
7	Forgetting bracing and lateral load requirements	2% of total dead load minimum
8	Misreading form removal criteria	70% design strength for slabs/beams
9	Overlooking OSHA requirements	Pre-erection, during, and post-placement inspections
10	Using 150 pcf for falsework concrete	160 pcf for falsework; 150 pcf for formwork
11	Forgetting 100% reshoring plan	8 reshores per 10 shores removed
12	Neglecting temperature effects on curing	Cold weather → slower strength gain → longer reshore
13	Using wrong CD factor for wood shoring	CD = 1.25 for ≤7-day construction loads
14	Forgetting minimum lateral bracing	100 lb/ft or 2% of vertical load
15	Misunderstanding backshoring vs. reshoring	Backshoring: sequential; Reshoring: after full removal

9. Practice Problems with Detailed Solutions

Problem 1 – Maximum Shore Load

Problem: A 12-in. thick slab is placed on a two-shore, one-reshore system. Slab weight D = 150 psf. What is the maximum shore/reshore load?

Solution:

1. Maximum shore/reshore load = **1.5D** [66]
2. $1.5 \times 150 = 225$ psf
3. **Answer: 225 psf**

Problem 2 – Reshore Load Distribution

Problem: During reshoring, the total load to be distributed is 300 psf. What is the load on the top reshore and the lower reshore?

Solution:

1. Top reshore = 2/3 of total load
2. Top reshore = $2/3 \times 300 = 200$ psf
3. Lower reshore = $1/3 \times 300 = 100$ psf
4. **Answer: Top = 200 psf; Lower = 100 psf**

Problem 3 – Shoring Design Check

Problem: A wood shore (6×6, A = 30.25 in², Fc = 1,200 psi, CD = 1.25, Cp = 0.7) supports a tributary load of 15,000 lb. Is the shore adequate?

Solution:

1. Adjusted allowable stress: $F_c' = 1,200 \times 1.25 \times 0.7 = 1,050$ psi
2. Allowable load: $P = 1,050 \times 30.25 = 31,763$ lb
3. 31,763 lb > 15,000 lb ✓ **Adequate**

Problem 4 – Form Removal Criteria

Problem: A slab is designed for f'c = 4,000 psi. At form removal, what is the minimum required concrete strength?

Solution:

1. Minimum strength = **70% of specified 28-day strength**
2. $0.70 \times 4,000 = 2,800$ psi
3. **Answer: 2,800 psi**

Problem 5 – OSHA Written Design Requirement

Problem: A contractor proposes tiered single post shores for a multistory project. What OSHA requirement applies?

Solution:

1. **Written design** by a qualified designer required [16]
2. **Inspection** by an engineer qualified in structural design
3. Additional requirements: vertical alignment, splicing, bracing in two perpendicular directions
4. **Answer: Written design and inspection by qualified engineer required**

Problem 6 – 100% Reshoring Plan

Problem: A contractor removes 20 shores from a slab. How many reshores must be installed under a 100% reshoring plan?

Solution:

1. 100% reshoring plan = **8 reshores per 10 shores removed**
2. $20 \text{ shores} \times (8/10) = 16$ reshores
3. **Answer: 16 reshores**

10. High-Yield Checklist

Definitions

□ Shoring: vertical/inclined supports during concrete placement

□ Reshoring: after form removal, distribute loads across floors

□ Backshoring: sequential, no deflection allowed

Load Path

□ Fresh concrete → shores → floor below → reshores → next floor

□ Maximum shore/reshore load = **1.5D**

□ Top reshore = **2/3** of load; lower reshore = **1/3**

□ 100% reshoring plan = **8 reshores per 10 shores removed**

Design Loads

□ Minimum combined vertical = **100 psf** (125 psf for motorized carts)

□ Live load minimum = **50 psf** (75 psf for carts)

□ Lateral bracing = **100 lb/ft** or **2% of vertical load**

□ Wind minimum = **15 psf**

Design

□ Axial capacity check for shoring members

□ Buckling check for long shores

□ NDS CD = **1.25** for construction loads

□ Slab capacity check for reshoring

OSHA Requirements

□ Drawings at jobsite

□ Pre-erection inspection

□ Inspection before, during, and after placement

□ Written design for **tiered single post shores**

□ Reshoring when concrete must support excess loads

Form Removal

□ Walls/columns: **2× stress** rule of thumb

□ Slabs/beams: **70% of design strength**

□ Field-cured cylinders required

□ Reshores remain until 28-day strength

11. Self-Test

1. What is the maximum shore/reshore load in a two-shore, one-reshore system?

- A) 1.0D
- B) 1.2D
- C) 1.5D
- D) 2.0D

Answer: C) 1.5D

2. What is the minimum combined vertical design load for shoring with hand tools?

- A) 50 psf
- B) 75 psf
- C) 100 psf
- D) 125 psf

Answer: C) 100 psf

3. When is a written design by a qualified designer required for shoring?

- A) All shoring operations
- B) Tiered single post shores

- C) Shoring over 16 ft
- D) Never

Answer: B) Tiered single post shores

4. What is the minimum concrete strength for form removal from slabs?

- A) 50% of design strength
- B) 70% of design strength
- C) 85% of design strength
- D) 100% of design strength

Answer: B) 70% of design strength

5. What is the load on the top reshore in a reshoring system?

- A) 1/3 of total load
- B) 1/2 of total load
- C) 2/3 of total load
- D) Full load

Answer: C) 2/3 of total load

6. How many reshores are required under a 100% reshoring plan when 30 shores are removed?

- A) 15
- B) 20
- C) 24
- D) 30

Answer: C) 24 (8 per 10 shores removed)

7. What is the minimum lateral bracing load for shoring?

- A) 1% of vertical load
- B) 2% of vertical load
- C) 3% of vertical load

- D) 5% of vertical load

Answer: B) 2% of vertical load (or 100 lb/ft, whichever is greater)

8. What is the NDS load duration factor (CD) for construction loads lasting ≤7 days?

- A) 1.0
- B) 1.15
- C) 1.25
- D) 1.6

Answer: C) 1.25

9. When must reshoring be erected?

- A) Before formwork removal
- B) As original forms and shores are removed
- C) 7 days after concrete placement
- D) At 28-day strength

Answer: B) As original forms and shores are removed

10. What is the minimum wind design load for shoring?

- A) 5 psf
- B) 10 psf
- C) 15 psf
- D) 20 psf

Answer: C) 15 psf

12. Flagged Items – Outside NCEES Scope or Requiring Verification

The following items could not be fully verified against NCEES-published specifications or the PE Civil Reference Handbook, Version 2.0:

1. **Exact NDS adjustment factors for shoring:** CD = 1.25 is confirmed for construction loads ≤7 days. Other adjustment factors (CM, Ct, CF, Ci) should be verified against the NDS Supplement for specific wood species and conditions.
2. **ASCE 37 construction live load values:** The range of 20–70 psf is confirmed, but specific values for shoring design should be verified against ASCE 37 tables and the PE Civil Reference Handbook.
3. **ACI 347.2R Table 5.5 exact values:** The simplified load distribution table is referenced, but exact values should be verified against the current edition of ACI 347.2R-17.
4. **OSHA 1926 Subpart P excavation shoring tables:** Timber shoring tables (Appendix C) and aluminum hydraulic shoring (Appendix D) are referenced. Specific values should be verified against the current OSHA regulations.
5. **SI unit equivalents:** SI conversions (kPa, kN) should be verified against the PE Civil Reference Handbook for exam problems requiring SI units.
6. **Post-tensioned construction load redistribution:** ACI 347.2R Chapter 4.7 covers post-tensioning load redistribution, but specific provisions may be outside the typical exam scope.

This review is designed to maximize readiness for the NCEES PE Civil: Construction exam. It does not guarantee a passing score. Always verify formulas and provisions against the current PE Civil Reference Handbook and referenced standards during the exam.

138 web pages

Act as a licensed PE an...
TXT 6.1KB

Thought for 4 seconds

> The user wants a comprehensive review of Bracing and Anchorage for the NCEES PE Civil: Construction exam. I need to gather authoritative information on design loads, stability checks, OSHA requirements, and code provisions. I will search for relevant sources including ACI 347R, ASCE 37, OSHA regulations, and NDS/AISC provisions. I'll also look for information on wind loads on shoring towers, formwork bracing, and anchorage design. I'll perform multiple searches to cover these topics.

Found 51 web pages

The search results provide some relevant information. I need to open specific pages to extract more detailed content. I'll open the ACI 347R bracing requirements, the OSHA shoring requirements, the wind load shape factor details, and the NDS wood bracing design information.

Read 4 pages

(a) Design: The design of falsework and formwork shall conform to the AASHTO Guide Design Specifications for ...

1926.703 - Requirements for cast-in-place concrete. - - Part Number:1926

Appendix D Example 5 – Horizontal Design Load on Heavy Duty Falsework

C15.2.3.2 The slenderness ratio \((\ell_{\mathrm{i}} / \mathrm{d}_{\mathrm{i}})\) limit of 80 for the individual me...

The opened pages provide some useful information. I need to gather more details on ASCE 37 load combinations, anchorage design, and connection design. I'll perform additional searches.

Found 50 web pages

The search results provide a good foundation for the review. I need to structure the response to cover all the subtopics requested by the user. I'll organize it into sections: Purpose and Principles, Types of Bracing, Types of Anchorage, Design Loads, Stability Checks and Design Procedures, OSHA and Code Requirements, Construction Applications and Quality Control, Common Errors and Exam Traps, Worked Examples, Practice Problems, High-Yield Checklist, and Self-Test. I'll incorporate the information from the search results, citing the relevant sources. I'll also flag any items that require verification.

Bracing and Anchorage for the NCEES PE Civil: Construction Exam

Exam-Focused Review | Design for Support of Construction Loads

Scope Note: This review covers only **Bracing and Anchorage** within the "Design for Support of Construction Loads" area. Formwork, falsework, shoring, and scaffolding are referenced only where necessary to explain bracing and anchorage behavior. All content is aligned with the NCEES PE Civil: Construction CBT Exam Specifications and the PE Civil Reference Handbook, Version 2.0.

1. Purpose and Principles

1.1 Role of Bracing and Anchorage in Temporary Structures

Bracing and anchorage systems are the critical elements that prevent temporary structures from failing due to lateral instability. Their primary functions are:

Function	Description	Typical Application
Lateral stability	Resist horizontal forces (wind, equipment, construction sequence)	Formwork, falsework, shoring towers
Overturning resistance	Prevent tipping due to eccentric or lateral loads	Shoring towers, wall forms
Sliding resistance	Prevent horizontal movement at the base	Scaffolding, shoring
Uplift resistance	Resist vertical forces that could lift the structure	Formwork subjected to wind uplift
Buckling prevention	Reduce effective length of compression members	Shoring posts, bracing members
Alignment maintenance	Keep forms and shores plumb and true	All temporary structures

1.2 Stability Concepts

Concept	Definition	Failure Mode
Overturning	Rotation about a pivot point due to lateral load exceeding restoring moment	Tower tips over
Sliding	Horizontal displacement at the base when lateral force exceeds friction + anchorage resistance	Structure slides
Uplift	Vertical force (typically wind) that exceeds the structure's weight	Structure lifts off foundation

Concept	Definition	Failure Mode
Lateral displacement	Excessive horizontal movement that causes misalignment or secondary stresses	Serviceability failure
Buckling	Sudden failure of a compression member due to instability	Bracing or shore member fails

1.3 Load Path from Applied Loads to Bracing and Anchorage

The typical load path for lateral loads in temporary structures:

1. **Applied lateral load** (wind, equipment, construction sequence) acts on the structure
2. **Diaphragm action** or **direct contact** transfers load to vertical bracing systems
3. **Vertical bracing** (diagonal, cross, or knee bracing) transfers load to the foundation or anchorage points
4. **Anchorage system** (tie-downs, anchor bolts, guy wires) resists the reaction at the base

Key principle: Every lateral load must have a continuous load path to a competent anchorage point. Any break in this path leads to instability.

1.4 Common Errors and Exam Traps

Error	Correct Approach
Assuming gravity loads govern all temporary structures	Lateral loads often govern bracing and anchorage design
Neglecting construction sequence loads	Unbalanced concrete pressure and equipment movements create lateral forces
Forgetting that bracing must work in **two perpendicular directions**	Single-direction bracing allows instability in the orthogonal direction
Overlooking uplift from wind	Wind can create uplift on formwork and towers

2. Types of Bracing

2.1 Lateral Bracing

Lateral bracing resists horizontal forces perpendicular to the main structural system.

Characteristics:

- Typically consists of diagonal members forming triangulated systems
- Must be provided in **both horizontal and vertical planes**
- Designed for tension, compression, or combined forces

2.2 Diagonal Bracing

Diagonal bracing members act in **tension and compression** to transfer lateral loads.

Design considerations:

- **Tension-only bracing:** Members resist only tension; compression member buckles and is ignored
- **Compression bracing:** Members must be designed for buckling (slenderness ratio limits)
- **Effective length factors (K):** Typically 0.65–1.0 for braced frames (pinned connections)

2.3 Cross Bracing

Cross bracing consists of two diagonal members crossing each other (X-bracing).

Key points:

- Provides stability in both directions
- One member acts in tension, the other in compression (or both tension in tension-only systems)
- Commonly used in shoring towers and falsework bents

2.4 Knee Bracing

Knee bracing connects beams to columns at their intersection to resist lateral forces.

Characteristics:

- Transfers moment through the connection
- Used when full diagonal bracing is not feasible
- Must be designed for combined bending and axial forces

2.5 Portal Bracing

Portal bracing creates a rigid frame opening for access through a braced structure.

Key points:

- Used where diagonal bracing would block a passage
- Designed for moment resistance at the connection
- Common in bridge falsework and shoring towers

2.6 Plan Bracing

Plan bracing is horizontal bracing that stabilizes a structure in the horizontal plane.

Applications:

- Stabilizes the tops of shoring towers
- Distributes lateral loads to vertical bracing systems
- Prevents racking of rectangular frames

2.7 Bracing for Specific Temporary Structures

Structure	Bracing Type	Key Requirement
Formwork (walls)	Diagonal braces	Minimum 100 lb/ft of wall applied at top
Falsework	Cross bracing + plan bracing	Minimum 2% of total dead load in any direction

Structure	Bracing Type	Key Requirement
Shoring towers	Diagonal + plan bracing	Two mutually perpendicular directions at splice level (tiered shores)
Scaffolding	Guys, ties, braces	Tie-in required when height-to-base ratio > 4:1
Excavation support	Cross bracing (trench jacks)	Per OSHA Subpart P requirements

2.8 Temporary and Permanent Bracing

Aspect	Temporary Bracing	Permanent Bracing
Duration	During construction only	Life of structure
Governing code	ASCE 37, OSHA	ASCE 7, IBC
Safety factor	Typically lower (construction exposure is shorter)	Design-level
Design live load	Reduced (ASCE 37 permits reductions)	Full design live load

2.9 Common Errors and Exam Traps – Bracing Types

Error	Correct Approach
Using only tension bracing without checking compression	If compression members exist, they must be designed for buckling
Forgetting plan bracing for shoring towers	Plan bracing stabilizes the top of towers
Missing bracing in one perpendicular direction	Two directions always required
Using permanent bracing loads for temporary bracing	ASCE 37 permits reduced construction loads

3. Types of Anchorage

3.1 Tie-Downs

Tie-downs resist **uplift** forces on formwork and temporary structures.

Common types:

- Screw anchors into concrete
- Embedded anchors cast into concrete
- Soil anchors (helical or grouted)

Design check:

$$T_{\text{required}} = \text{Uplift force} - \text{Dead load resisting}$$

$$T_{\text{capacity}} \geq T_{\text{required}} \times FS$$

Typical FS for tie-downs: **2.0** for form ties (per some specifications).

3.2 Anchor Bolts

Anchor bolts resist **tension, shear, or combined** loading at connections.

Design provisions (ACI 318 Chapter 17):

- Steel strength in tension: $\phi N_{sa} = \phi A_{se} f_{uta}$
- Concrete breakout in tension: ϕN_{cbg}
- Pullout strength: ϕN_{pn}
- Steel strength in shear: ϕV_{sa}
- Concrete breakout in shear: ϕV_{cbg}
- **Combined tension and shear interaction:**

$$\left(\frac{N_u}{\phi N_n}\right)^{5/3} + \left(\frac{V_u}{\phi V_n}\right)^{5/3} \leq 1.0$$

3.3 Embedded Anchors and Cast-in-Place Anchors

Anchor Type	Description	Typical Use
Cast-in-place	Installed before concrete placement	Formwork tie-downs, crane anchors
Post-installed	Drilled and set after concrete hardens	Retrofit bracing, temporary tie-downs
Mechanical	Expansion or undercut anchors	Scaffolding tie-ins
Adhesive	Chemical bonding	High-load applications

Critical exam point: Anchorage design must consider **concrete strength at time of loading** — early-age concrete has lower capacity.

3.4 Deadmen, Guy Wires, and Soil/Rock Anchors

System	Description	Application
Deadman	Buried anchor block that resists horizontal force through soil resistance	Excavation bracing, guy wire anchorage
Guy wire	Tension cable that transfers lateral load to a distant anchor point	Shoring towers, scaffolding
Soil anchor	Grouted tendon or helical anchor installed in soil	Retaining walls, excavation support
Rock anchor	Grouted bar or strand installed in rock	Rock excavation support

3.5 Anchorage to Concrete, Steel, and Soil

Substrate	Key Design Considerations
Concrete	Embedment depth, edge distance, concrete strength, cracked vs. uncracked
Steel	Bolt shear, weld strength, base metal thickness

Substrate	Key Design Considerations
Soil	Soil type, bearing capacity, pullout resistance, corrosion protection

3.6 Connection Design: Bolts, Welds, and Embedments

Bolted connections (AISC):

- Bolt shear: $V = F_v A_b$ (ASD)
- Bolt bearing: $V = 2.4 d t F_u$ (ASD)
- Bolt tension: $T = F_t A_b$
- Combined shear + tension: interaction equation

Welded connections (AISC):

- Fillet weld strength: $\phi R_n = \phi (0.60 F_{exx})(0.707 w) L$
- Minimum weld size per material thickness
- Effective length requirements

Embedments:

- Development length for reinforcing bars
- Hook and headed stud capacity
- Concrete breakout cone

3.7 Common Errors and Exam Traps – Anchorage

Error	Correct Approach
Neglecting concrete breakout for anchors	Check all limit states: steel, breakout, pullout, side-face blowout
Using full concrete strength for anchors loaded early	Use concrete strength at time of loading
Forgetting edge distance requirements	Edge distance affects breakout capacity

Error	Correct Approach
Assuming friction contributes to sliding resistance	Friction is typically ignored for temporary structures
Using wrong FS for tie-downs	Typically 2.0 for form ties; verify specification

4. Design Loads for Bracing and Anchorage

4.1 Lateral Loads

Load Type	Source	Typical Value
Wind	ASCE 37, local codes	Minimum 15 psf; higher for exposed towers
Seismic	ASCE 37	Reduced for short construction duration
Construction sequence	Unbalanced concrete, equipment	Varies
Equipment	Cranes, concrete pumps, buggies	As specified
Unbalanced concrete pressure	Formwork	Calculated from ACI 347R

4.2 Minimum Lateral Load Requirements

ACI 347R / AASHTO requirements:

$$H_{\min} = \max\left(100 \text{ lb/ft along slab edge},\ 2\% \text{ of supported vertical load}\right)$$

For **wall forms:** Minimum **100 lb/ft of wall** applied at the top.

For **falsework bracing:** The horizontal load shall be the sum of all lateral loads, but **in no case less than 2% of the total dead load** in any direction. [1]

ASCE 37: The 2% lateral load **need not be applied concurrently with wind or seismic load**.

4.3 Minimum Wind Design Load

- **Formwork:** Minimum **15 psf** (per ACI 347R)
- **Falsework:** Per AASHTO and state DOT specifications
- **Heavy-duty steel shoring towers:** Shape factor **2.2** applied to height-zone wind pressures

4.4 Wind Loads on Heavy-Duty Shoring Towers

Shape factor: 2.2 for heavy-duty steel shoring.

Height zone wind pressure table (Caltrans example):

Height Zone (ft)	Wind Pressure (psf)	Design Wind Load (psf)
0–30	20	20 × 2.2 = 44
30–50	25	25 × 2.2 = 55
50–100	30	30 × 2.2 = 66
100–105	35	35 × 2.2 = 77

Conventional falsework drag coefficient: $Q = 1 + 0.2W$, capped at 10. [3][2]

4.5 Load Combinations for Bracing and Anchorage Design

Per ASCE 37, typical load combinations for temporary structures:

1. **Dead + Construction Live + Lateral** (governing for most bracing)
2. **Dead + Wind** (wind often governs for tall, exposed towers)
3. **Dead + Seismic** (reduced seismic for short duration)
4. **Dead + Construction Lateral** (equipment, sequence)

ASD load combinations: Use service loads with allowable stress design.

4.6 Common Errors and Exam Traps – Design Loads

Error	Correct Approach
Using 15 psf when higher wind is specified	Check local codes and ASCE 37
Forgetting the 100 lb/ft minimum for wall forms	Apply at the top of wall forms
Applying 2% lateral load concurrently with wind	Per ASCE 37, they need not be concurrent
Using shape factor 1.0 for shoring towers	Shape factor = 2.2 for heavy-duty steel shoring
Neglecting construction sequence loads	Include unbalanced concrete pressure

5. Stability Checks and Design Procedures

5.1 Overturning Stability

Factor of safety against overturning:

$$FS = \frac{M_{\text{resisting}}}{M_{\text{overturning}}} \geq 1.25$$

Step-by-step procedure:

1. Calculate overturning moment about the critical pivot point (typically the toe of the base)
2. Calculate resisting moment from dead load of the structure
3. If FS < 1.25, provide additional anchorage (tie-downs, guy wires)
4. Design anchorage for the net overturning moment

Example: A shoring tower has an overturning moment of 100,000 ft-lb. Dead load resisting moment = 80,000 ft-lb.

- FS = 80,000 / 100,000 = 0.80 < 1.25 → **Not stable**

- Required anchorage = 1.25 × 100,000 − 80,000 = 45,000 ft-lb

5.2 Sliding Resistance and Anchorage Capacity

Sliding resistance:

$$V_{\text{resisting}} = V_{\text{friction}} + V_{\text{anchorage}}$$

Note: Friction is typically **ignored** for temporary structures due to uncertainty.

Anchorage capacity: Sum of all anchors' shear capacities ≥ applied lateral load × FS.

5.3 Uplift Resistance and Tie-Down Design

$$T_{\text{required}} = W_{\text{dead}} \times \text{safety factor} - W_{\text{dead}}$$

Or more directly: $T_{\text{required}} = \text{Uplift force} \times FS$

Typical FS for tie-downs: 2.0 (per some specifications for form ties).

5.4 Lateral Displacement and Drift Limits

- **Formwork:** Typically limited to l/360 for sheathing, l/240 for studs/wales
- **Scaffolding:** 1/60 of span for platform deflection
- **Shoring towers:** Engineer-specified; often H/500

5.5 Buckling of Bracing Members

Euler buckling equation:

$$P_{\text{cr}} = \frac{\pi^2 EI}{(KL)^2}$$

Slenderness ratio:

$$\lambda = \frac{KL}{r}$$

Effective length factors (K):

End Condition	K
Pinned–pinned	1.0
Fixed–pinned	0.8
Fixed–fixed	0.65
Fixed–free (cantilever)	2.1

NDS requirements for wood bracing:

- Maximum slenderness ratio for solid columns: **50** (l/d)
- Spaced columns: individual member limit **80**; l_3/d_1 limit **40** [23]

5.6 Design of Bracing Members

Tension members:

$$f_t = \frac{T}{A_n} \leq F'_t$$

Compression members:

$$f_c = \frac{P}{A} \leq F'_c \quad (\text{with } C_p \text{ adjustment})$$

Combined bending + axial (NDS interaction):

$$\left(\frac{f_c}{F'_c}\right)^2 + \frac{f_b}{F'_b} \leq 1.0$$

5.7 Design of Anchorage for Tension, Shear, and Combined Loading

ACI 318 Chapter 17 provisions:

Limit State	Design Equation
Steel tension	$\phi N_{sa} = \phi A_{se} f_{uta}$

Limit State	Design Equation
Concrete breakout tension	ϕN_{cbg}
Steel shear	$\phi V_{sa} = \phi 0.6 A_{se} f_{uta}$
Concrete breakout shear	ϕV_{cbg}
Combined	$(N_u/\phi N_n)^{5/3} + (V_u/\phi V_n)^{5/3} \leq 1.0$

5.8 Step-by-Step Design Procedure for Bracing and Anchorage

Step 1 – Determine lateral loads

- Wind, seismic, equipment, construction sequence
- Apply minimum lateral load requirement

Step 2 – Establish load path

- Identify how lateral loads transfer to bracing systems
- Identify anchorage points

Step 3 – Design bracing members

- Tension: check net area
- Compression: check buckling with effective length
- Combined: interaction equation

Step 4 – Check overturning stability

- FS ≥ 1.25
- If not, provide anchorage

Step 5 – Design anchorage

- Tension, shear, combined
- Check concrete breakout, pullout, steel strength

Step 6 – Check connections

- Bolts, welds, embedments
- Per AISC, NDS, or ACI 318

Step 7 – Verify lateral displacement

- Check drift limits

5.9 Common Errors and Exam Traps – Stability and Design

Error	Correct Approach
Using FS = 1.0 for overturning	FS = **1.25** required
Including friction in sliding resistance	Friction is typically ignored
Using K = 1.0 for all columns	Use appropriate K based on end conditions
Forgetting buckling check for compression bracing	Always check slenderness ratio
Missing combined tension + shear check for anchors	Use ACI 318 interaction equation

6. OSHA and Code Requirements

6.1 OSHA 1926 Subpart Q – Concrete and Masonry Construction

Section	Requirement
1926.703(a)(1)	Formwork must support all vertical and lateral loads without failure [17]
1926.703(a)(2)	Drawings/plans available at jobsite [17]
1926.703(b)(1)	Shoring equipment inspected prior to erection [17]

Section	Requirement
1926.703(b)(3)	Erected shoring inspected before, during, and after placement [17]
1926.703(b)(8)	**Tiered single post shores:** written design by qualified designer, inspected by structural engineer [17]
1926.703(b)(8)(iv)	Tiered shores braced in **two mutually perpendicular directions** at splice level; each tier diagonally braced [17]

Written design requirement: OSHA 1926.703(b)(8)(i) requires written design for **tiered single post shores** (not all shoring). However, drawings/plans for all formwork and shoring must be available at the jobsite.

6.2 OSHA 1926 Subpart L – Scaffolding

Requirement	Specification
Tie-in trigger	Height-to-base ratio **> 4:1**
Top tie	Placed no further than **4:1 ratio from the top**
Vertical tie spacing	Repeated at **26 ft intervals** thereafter
Bracing	Guys, ties, braces in two perpendicular directions
Platform capacity	**4:1** safety factor

6.3 OSHA 1926 Subpart P – Excavations

- **Protective systems** required for excavations **5 ft or deeper**
- **Design by PE** required for excavations **20 ft or deeper**
- **Timber shoring tables:** Appendix C
- **Aluminum hydraulic shoring:** Appendix D
- **Support adjacent structures:** Shoring, bracing, or underpinning required

6.4 ASCE 37 – Design Loads on Structures During Construction

- Provides load combinations for temporary structures
- Construction live loads: 20–70 psf
- Wind load reductions permitted for short construction exposure
- Seismic load reductions permitted for short duration
- **2% lateral load need not be concurrent with wind/seismic**

6.5 ACI 347R – Formwork Bracing and Anchorage

- Minimum lateral load: **100 lb/ft** along slab edge or **2% of vertical load**, whichever is greater
- Wall form bracing: **100 lb/ft** applied at top
- Minimum wind: **15 psf**

6.6 AISC and NDS Provisions

Standard	Application
AISC 360	Steel bracing member design, connection design
AISC Design Guide 29	Vertical bracing connections
NDS	Wood bracing member design, column stability factor
NDS Supplement	Reference design values for wood species/grades

6.7 Common Errors – OSHA and Code

Error	Correct Approach
Forgetting written design for tiered shores	Required by OSHA 1926.703(b)(8)(i)
Missing two-direction bracing for tiered shores	Required at splice level and each tier
Using 2% lateral load concurrently with wind	ASCE 37 permits them to be separate

Error	Correct Approach
Forgetting tie-in at 4:1 height-to-base ratio	Required for supported scaffolds
Neglecting OSHA Subpart P for excavation bracing	Protective systems required at 5 ft

7. Construction Applications and Quality Control

7.1 Bracing and Anchorage for Multistory Construction

- **Shoring towers** must be braced in two perpendicular directions
- **Tiered shores** require special bracing at splice level
- **Reshoring** requires anchorage to prevent lateral displacement
- **Slab edges** must be braced for 100 lb/ft or 2% of load

7.2 Bracing and Anchorage for Bridges and Special Structures

- **Falsework bents** require cross bracing and plan bracing
- **Heavy-duty shoring towers** designed for shape factor 2.2 wind
- **Overhang brackets** require tie-down anchorage
- **Cantilever falsework** requires counterweight or anchorage

7.3 Bracing and Anchorage for Excavation Support

- **Sheet piling** requires walers and struts
- **Soldier piles** require lagging and bracing
- **Trench boxes** provide pre-engineered bracing
- **Tiebacks** anchor to soil or rock behind the excavation

7.4 Inspection, Tolerances, and Specification Conformance

- **Pre-erection inspection** of all bracing and anchorage components

- **During placement inspection** – immediately prior to, during, and after
- **Tolerances:** Plumbness, alignment, and elevation within specification
- **Documentation:** Drawings and plans available at jobsite

7.5 Documentation and Quality Control

- Bracing and anchorage installed per authorized shop drawings
- Components inspected prior to erection
- Independent design check may be required
- As-built inspection by engineer in responsible charge

7.6 Common Field and Design Errors

Error	Consequence
Missing bracing in one direction	Lateral instability and collapse
Inadequate anchorage capacity	Overturning or sliding failure
Using damaged bracing components	Reduced capacity
Improper vertical alignment	Eccentric loading and buckling
Insufficient tie-downs	Uplift failure during wind

8. Common Errors and Exam Traps – Master List

#	Trap	Correct Approach
1	Unit inconsistencies (USCS vs. SI)	Check units in every formula
2	Confusing bracing and anchorage definitions	Bracing resists forces; anchorage connects to support

#	Trap	Correct Approach
3	Forgetting minimum lateral load (100 lb/ft or 2%)	Apply the **greater** of the two
4	Neglecting overturning FS = 1.25	Required in all directions
5	Misunderstanding wind load tables and shape factors	Shape factor = 2.2 for heavy-duty steel shoring
6	Overlooking OSHA written design for tiered shores	Required by OSHA 1926.703(b)(8)(i)
7	Forgetting tie-in at 4:1 height-to-base ratio	Required for supported scaffolds
8	Misreading connection design capacities	Check bolt shear, weld strength, embedment
9	Neglecting buckling checks for bracing members	Always check slenderness ratio
10	Overlooking anchorage capacity for uplift and sliding	Check tension, shear, combined
11	Including friction in sliding resistance	Friction is typically ignored
12	Using K = 1.0 for all columns	Use appropriate K based on end conditions
13	Forgetting concrete breakout for anchors	Check all ACI 318 Chapter 17 limit states
14	Applying 2% lateral load concurrently with wind	ASCE 37 permits separate application
15	Missing two-direction bracing requirement	Required for all temporary structures

9. Practice Problems with Detailed Solutions

Problem 1 – Minimum Lateral Load

Problem: A wall form is 20 ft long and 12 ft high. The total vertical load supported is 10,000 lb. What is the minimum lateral design load for bracing?

Solution:

1. 100 lb/ft × 20 ft = 2,000 lb
2. 2% × 10,000 = 200 lb
3. Greater = 2,000 lb
4. **Answer: 2,000 lb** (100 lb/ft governs)

Problem 2 – Overturning Stability

Problem: A shoring tower has an overturning moment of 150,000 ft-lb. The resisting moment from dead load is 120,000 ft-lb. Is the tower stable? If not, what anchorage moment is required?

Solution:

1. FS = 120,000 / 150,000 = 0.80
2. Required FS = 1.25
3. Required resisting moment = 1.25 × 150,000 = 187,500 ft-lb
4. Required anchorage = 187,500 − 120,000 = 67,500 ft-lb
5. **Answer: Not stable; requires 67,500 ft-lb anchorage**

Problem 3 – Buckling of Bracing Member

Problem: A wood brace (4×4, A = 12.25 in², r = 0.98 in, E = 1,700,000 psi) is 8 ft long with pinned ends (K = 1.0). What is the Euler critical load?

Solution:

1. KL = 1.0 × 8 × 12 = 96 in
2. Slenderness ratio = 96 / 0.98 = 98

3. $P_{\text{cr}} = \pi^2 \times 1,700,000 \times 12.5/(96)^2$
4. $P_{\text{cr}} = 209,800,000/9,216 = 22,764$ lb
5. **Answer: 22,764 lb** (Euler critical load)

Problem 4 – Wind Load on Shoring Tower

Problem: A heavy-duty steel shoring tower is 60 ft tall in the 50–100 ft height zone (wind pressure = 30 psf). The projected area is 2.0 ft²/ft. What is the design wind load per foot?

Solution:

1. Shape factor = 2.2
2. Design wind pressure = 30 × 2.2 = 66 psf
3. Wind load per foot = 66 × 2.0 = 132 lb/ft
4. **Answer: 132 lb/ft**

Problem 5 – Anchor Combined Loading

Problem: An anchor is subjected to tension $N_u = 5,000$ lb and shear $V_u = 3,000$ lb. The design capacities are $\phi N_n = 8,000$ lb and $\phi V_n = 6,000$ lb. Check the interaction.

Solution:

1. $(5,000/8,000)^{5/3} + (3,000/6,000)^{5/3}$
2. $(0.625)^{5/3} + (0.5)^{5/3}$
3. $0.456 + 0.315 = 0.771$
4. 0.771 ≤ 1.0 → **Adequate**

Problem 6 – Scaffolding Tie-In

Problem: A supported scaffold is 30 ft tall with a base width of 6 ft. Is a tie-in required?

Solution:

1. Height-to-base ratio = 30 / 6 = 5.0
2. 5.0 > 4.0 → **Tie-in required**

3. Top tie must be placed no further than 4:1 ratio from top (24 ft from top)
4. **Answer: Tie-in required**

10. High-Yield Checklist

Purpose and Principles

□ Bracing resists lateral forces; anchorage connects to support

□ Load path must be continuous to a competent anchorage

□ Stability concepts: overturning, sliding, uplift, buckling

Types of Bracing

□ Lateral, diagonal, cross, knee, portal, plan bracing

□ Two perpendicular directions always required

□ Tiered shores: braced at splice level and each tier diagonally

Types of Anchorage

□ Tie-downs resist uplift

□ Anchor bolts resist tension, shear, combined

□ Deadmen, guy wires, soil/rock anchors

□ ACI 318 Chapter 17 for concrete anchorage

Design Loads

□ Minimum lateral load = 100 lb/ft or 2% of vertical load

□ Wall forms: 100 lb/ft at top

□ Minimum wind = 15 psf

□ Heavy-duty shoring shape factor = 2.2

Stability Checks

□ Overturning FS = **1.25** minimum

□ Friction typically ignored for sliding

□ Buckling check for all compression members

□ Slenderness ratio limits: 50 for solid columns (NDS)

OSHA and Codes

□ Tiered shores: written design by qualified designer

□ Tiered shores: braced in two perpendicular directions

□ Scaffolding: tie-in at > 4:1 height-to-base ratio

□ Excavations 5 ft+: protective systems required

Design Procedures

□ Check tension, compression, combined for bracing

□ Check steel, concrete breakout, pullout for anchors

□ Check bolt shear, weld strength for connections

□ Verify lateral displacement limits

11. Self-Test

1. What is the minimum lateral load for bracing a wall form?

- A) 50 lb/ft
- B) 100 lb/ft
- C) 200 lb/ft
- D) 2% of vertical load

Answer: B) 100 lb/ft (applied at top)

2. What is the minimum factor of safety against overturning for temporary structures?

- A) 1.0
- B) 1.25
- C) 1.5
- D) 2.0

Answer: B) 1.25

3. What is the shape factor for heavy-duty steel shoring towers?

- A) 1.0
- B) 1.5
- C) 2.0
- D) 2.2

Answer: D) 2.2

4. When is a written design by a qualified designer required for shoring?

- A) All shoring operations
- B) Tiered single post shores
- C) Shoring over 16 ft
- D) Never

Answer: B) Tiered single post shores

5. At what height-to-base ratio are tie-ins required for supported scaffolds?

- A) 2:1
- B) 3:1
- C) 4:1
- D) 6:1

Answer: C) 4:1

6. Which of the following is typically ignored in sliding resistance calculations?

- A) Dead load

- B) Friction
- C) Anchorage capacity
- D) Cohesion

Answer: B) Friction

7. What is the maximum slenderness ratio for a solid wood column per NDS?

- A) 30
- B) 40
- C) 50
- D) 80

Answer: C) 50

8. In the ACI 318 combined loading interaction equation, what is the exponent for tension and shear ratios?

- A) 1.0
- B) 1.5
- C) 5/3
- D) 2.0

Answer: C) 5/3

9. What is the minimum lateral load for falsework bracing?

- A) 1% of total dead load
- B) 2% of total dead load
- C) 3% of total dead load
- D) 5% of total dead load

Answer: B) 2% of total dead load

10. Which OSHA subpart governs excavation bracing?

- A) Subpart L

- B) Subpart P
- C) Subpart Q
- D) Subpart R

Answer: B) Subpart P

12. Flagged Items – Outside NCEES Scope or Requiring Verification

The following items could not be fully verified against NCEES-published specifications or the PE Civil Reference Handbook, Version 2.0:

1. **Exact NDS adjustment factors for bracing members:** CD = 1.25 for construction loads ≤7 days is confirmed. Other adjustment factors (CM, Ct, CF, Ci) should be verified against the NDS Supplement for specific wood species and conditions.
2. **ASCE 37 load combinations for temporary bracing:** ASCE 37 provides load combinations for construction, but specific combinations for bracing and anchorage should be verified against the standard and the PE Civil Reference Handbook.
3. **ACI 318 Chapter 17 anchor design provisions:** The interaction equation and limit states are summarized, but exact φ factors and edge distance modifiers should be verified against the current ACI 318 edition.
4. **State DOT wind pressure tables:** The Caltrans height zone table is provided as an example. Other states may use different values. The exam should specify which standard governs.
5. **SI unit equivalents:** SI conversions (kPa, kN, mm) should be verified against the PE Civil Reference Handbook for exam problems requiring SI units.
6. **Scaffolding tie spacing intervals:** The 26-ft vertical tie spacing is from OSHA non-mandatory Appendix E. Verify applicability to specific exam problems.

This review is designed to maximize readiness for the NCEES PE Civil: Construction exam. It does not guarantee a passing score. Always verify formulas and provisions against the current PE Civil Reference Handbook and referenced standards during the exam.

98 web pages

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Thought for 3 seconds

用户需要一份关于“Support of Excavation”的全面备考复习资料，这是PE Civil: Construction考试中“Design for Support of Construction Loads”部分的一个主题。我需要提供定义、公式、分步程序、例题、应试技巧、练习题和高收益检查清单，并引用PE Civil Reference Handbook Version 2.0和相关标准。为了全面获取所需信息，我将同时进行多项搜索，覆盖SOE系统、土压力、设计方法、稳定性、地下水控制、施工监测以及OSHA要求等关键领域。

Found 78 web pages

所有并行搜索已完成。现在需要根据搜索结果，整理一份关于“Support of Excavation”的全面备考复习资料。我将按照用户要求的格式，涵盖所有子主题，并引用搜索结果中的相关信息。为了获取更详细的内容，我需要同时打开一些相关的页面。

Read 8 pages 16.4.1 Notation

4.2. SHEET PILE WALLSCantilever sheet pile wall is a common type of temporary shoring system made of individ...

Design of Deep Excavations in Urban Environments

4. If \(\gamma \mathsf{H} / \mathsf{C} < 4\) , use section (b), If \(4 < \gamma \mathsf{H} / \mathsf{C} < 6\) , use...

6. Settlement of vertical wall elements can cause reduction of anchor loads and should be considered in design

View All

这些资料提供了关于土压力、支护系统设计、稳定性、地下水控制和OSHA要求的信息。现在我需要组织一份全面的复习资料。我的回答将涵盖所有子主题，包括定义、公式、分步程序、例题、应试技巧、练习题和高收益检查清单。我会引用搜索结果中的具体信息，并注明来源。现在开始组织回答。

Support of Excavation for the NCEES PE Civil: Construction Exam

Exam-Focused Review | Design for Support of Construction Loads

Scope Note: This review covers only **Support of Excavation** within the "Design for Support of Construction Loads" area. Formwork, falsework, shoring towers, and scaffolding are referenced only where necessary to explain excavation support behavior. All content is aligned with the NCEES PE Civil: Construction CBT Exam Specifications and the PE Civil Reference Handbook, Version 2.0.

Important: Support of Excavation is a **broad topic** that intersects geotechnical, structural, and construction engineering. This review prioritizes the most exam-relevant subtopics while flagging items that may require verification against NCEES-published specifications.

1. Excavation Support Systems and Selection

1.1 System Types and Characteristics

System	Description	Cost-Effective Height Range	Key Design Consideration
Soldier piles + lagging	Steel I-beams at 6–10 ft centers with timber/concrete lagging spanning between piles	Up to 15–18 ft (cantilever); >15 ft typically requires tiebacks	Soil arching increases effective width up to 3× hole diameter; lagging spans between piles [11]
Sheet pile walls (cantilever)	Interlocking steel sheets driven side-by-side forming continuous wall	Up to 16–18 ft	Large deflections possible; limited to shallow excavations [11]
Sheet pile walls (anchored)	Sheet piles with ground anchors or tiebacks	>15 ft	Anchors provide lateral resistance at depth
Braced excavations	Struts, walers, rakers, corner braces	Varies by depth	Internal bracing; struts in compression require buckling check
Anchored walls	Ground anchors, tiebacks, deadmen	>15 ft	Bond zone must extend beyond active failure wedge
Diaphragm/slurry walls	Cast-in-place concrete walls constructed using slurry trench	>30 ft	High cost; used where minimal deflection required
Secant/tangent pile walls	Interlocking or tangent bored piles	Varies	Used where groundwater control is critical
Soil nail walls	Passive inclusions installed in-situ	Varies	Top-down construction; requires global stability check

1.2 Selection Criteria

Selection depends on: **soil type** (granular vs. cohesive), **excavation depth**, **groundwater conditions**, **adjacent structures** (settlement tolerance), **site access**, **construction schedule**, and **cost** (initial + long-term performance).

1.3 Common Errors and Exam Traps

Error	Correct Approach
Using cantilever sheet piles for deep excavations	Cantilever limited to ~18 ft; deeper requires anchors/struts
Forgetting soil arching for soldier piles	Effective width up to 3× hole diameter in granular soils; **no arching** if stability number < 3 in soft clay [11]
Assuming lagging carries all lateral load	Lagging spans between piles; piles carry the load to anchors or embedment
Selecting SOE system without checking adjacent structures	Settlement behind wall can damage adjacent buildings

2. Lateral Earth Pressures and Loads

2.1 At-Rest, Active, and Passive Coefficients (Rankine)

Rankine Active Earth Pressure Coefficient:

$$K_a = \tan^2\left(45^\circ - \frac{\phi}{2}\right) = \frac{1-\sin\phi}{1+\sin\phi}$$

Rankine Passive Earth Pressure Coefficient:

$$K_p = \tan^2\left(45^\circ + \frac{\phi}{2}\right) = \frac{1+\sin\phi}{1-\sin\phi}$$

At-Rest (Jaky):

$$K_0 = 1 - \sin\phi \quad (\text{normally consolidated sands})$$

Where ϕ = effective angle of internal friction.

Coulomb vs. Rankine: Coulomb considers wall friction (δ) and is generally more conservative for active pressure (over-predicts Ka compared to Rankine). When δ = 0, β = 0, and α = 90°, both methods produce identical coefficients [21] .

2.2 Effect of Wall Friction, Backfill Slope, and Cohesion

- **Wall friction (δ):** Reduces active pressure; increases passive pressure
- **Backfill slope (β):** Increases active pressure
- **Cohesion (c):** Reduces active pressure; apparent earth pressure diagrams are used for cohesive soils
- **Tensile zone:** Rankine theory produces a "tensile" stress zone near the surface; depth of tension cracking should not exceed half the excavation height

2.3 Surcharge Loads

Surcharge Type	Pressure Distribution	Formula
Uniform (q)	Constant lateral pressure	$P_s = qK_a$
Point/line load	Boussinesq distribution	Per elastic theory
Strip load	Trapezoidal distribution	Per Boussinesq
Sloping ground	Additional active pressure	Per Rankine/Coulomb with β

Minimum construction surcharge: A minimum lateral pressure of **100 psf** for surcharge loads due to construction equipment or material stockpiles shall be used for design and distributed as a uniform pressure over the height of excavation . Heavy loads should be kept at least **15 ft** from the top of the excavation .

2.4 Water Pressure and Hydrostatic Effects

$$P_w = \gamma_w \times h_w$$

Where $\gamma_w = 62.4$ pcf (fresh water). Effective stress analysis must be used when groundwater is present:

$$\sigma'_v = \sigma_v - u$$

Critical: Water pressure must be added to earth pressure. Drainage systems (filters, weep holes) reduce hydrostatic pressure but must be designed to prevent clogging.

2.5 Seismic Earth Pressures (Mononobe-Okabe)

Active seismic coefficient:

$$K_{ae} = K_a + \Delta K_{ae}$$

Where ΔK_{ae} depends on horizontal acceleration coefficient k_h and vertical acceleration k_v. For most temporary excavations, seismic design is often **not required** due to short exposure duration, but must be checked for critical projects.

2.6 Load Combinations

Per service load design (ASD) for temporary excavation supports:

Wall System Type	Load Combination
Flexible walls	100% DL + LL + B + **80% of E** at 100% allowable stress
Semi-rigid walls	100% DL + LL + E + B at 100% allowable stress

All pressures are **additive**; no safety factors are included in earth pressure calculations themselves [31] .

2.7 Common Errors and Exam Traps – Earth Pressures

Error	Correct Approach
Using total stress for drained conditions	Use effective stress with ϕ' for granular soils
Forgetting water pressure	Add $P_w = \gamma_w h_w$ when groundwater present
Confusing active and passive coefficients	$K_a < K_0 < K_p$
Neglecting surcharge	Add $P_s = qK_a$ for all surcharges
Applying Coulomb when Rankine required	Use Rankine for simple wall geometry; Coulomb for wall friction
Forgetting tension crack depth	$z_c = \frac{2c}{\gamma\sqrt{K_a}}$; limit to H/2

3. Design of Excavation Support Systems

3.1 Structural Design of Soldier Piles and Sheet Piles

Soldier pile design (cantilever):

1. Determine active pressure diagram (triangular for granular, trapezoidal for cohesive)
2. Calculate maximum bending moment: $M_{\max}$
3. Determine required section modulus: $S_{req} = M_{\max}/F_b$
4. Determine embedment depth D for passive resistance
5. Check deflection

Effective width for passive resistance:

- Driven piles or drilled piles backfilled with non-concrete: use element width b
- Drilled piles with concrete: use **hole diameter**
- Soil arching may increase effective width up to **3×** hole diameter (AASHTO 3.11.5.6) [11]

3.2 Design of Walers, Struts, and Bracing

Walers: Horizontal members that distribute earth pressure from sheet piles/soldier piles to struts. Designed for bending.

Struts: Horizontal compression members spanning across the excavation.

- Design for axial compression + buckling
- $P_{\text{allow}} = F'_c \times A$ with column stability factor
- Maximum allowable working load of struts: **67% of mean failure load** (temporary excavations) or **50% of failure load** (permanent excavations)

Rakers: Inclined struts that transfer lateral load to the base slab or foundation. Design for combined axial + bending.

3.3 Design of Anchors and Tiebacks (FHWA GEC-4)

Components:

- **Unbonded length:** Minimum 3 m (bar) or 4 m (strand); must extend beyond active failure wedge
- **Bond length:** Develops anchor capacity through grout-ground bond
- **Maximum bonded length:** 12 m per FHWA (longer lengths cannot effectively transfer load)

Design capacity:

$$T_{\text{allow}} = \frac{T_{\text{ult}}}{FS}$$

Where FS typically ranges from **1.1 to 1.5** applied to shear strength .

Corrosion protection (FHWA GEC-4):

- **Class I:** Encapsulated in grout; minimum 0.50 in. grout cover
- **Class II:** Grout-protected; electrically isolated
- **Class III:** Simple grout protection

Internal and external failure modes must be checked per FHWA GEC-4 [41] .

3.4 Buckling and Effective Length Factors

End Condition	K
Pinned–pinned	1.0
Fixed–pinned	0.8
Fixed–fixed	0.65
Fixed–free	2.1

Slenderness ratio limit: For steel struts, $KL/r \leq 200$ (AISC preferred limit).

3.5 Connection Design

- **Bolted:** Shear, bearing, tension per AISC
- **Welded:** Fillet weld strength per AISC
- **Embedments:** Development length, hooks, headed studs per ACI 318

3.6 Step-by-Step Design Procedure

Step 1 – Evaluate site conditions

- Soil profile, groundwater, adjacent structures, surcharges

Step 2 – Determine earth pressure coefficients

- K_a, K_p, K_0 based on soil type and wall movement

Step 3 – Calculate lateral loads

- Earth pressure + water pressure + surcharge

Step 4 – Select SOE system

- Based on depth, soil, groundwater, adjacent structures

Step 5 – Design wall elements

- Soldier piles, sheet piles, or diaphragm walls

- Check bending, shear, deflection, embedment

Step 6 – Design bracing/anchors

- Struts: compression + buckling
- Anchors: bond length, unbonded length, capacity

Step 7 – Check stability

- Overturning, sliding, bearing, basal heave, piping

Step 8 – Check global stability

- Slope stability through and behind wall

Step 9 – Design connections

- Bolts, welds, embedments

Step 10 – Verify serviceability

- Deflection limits, settlement of adjacent structures

3.7 Common Errors – Design

Error	Correct Approach
Using triangular pressure for cohesive soils	Use trapezoidal or Peck empirical diagrams
Forgetting embedment depth check	Passive resistance must develop below excavation
Using full failure load for strut design	Limit to 67% (temporary) or 50% (permanent)
Neglecting anchor bond length requirements	Must extend beyond active wedge; max 12 m
Forgetting corrosion protection for permanent anchors	Required per FHWA GEC-4

4. Stability and Failure Modes

4.1 Overturning, Sliding, and Bearing Capacity

Overturning (cantilever walls):

$$FS_{\text{overturning}} = \frac{M_{\text{resisting}}}{M_{\text{overturning}}} \geq 1.5$$

Sliding:

$$FS_{\text{sliding}} = \frac{V_{\text{resisting}}}{V_{\text{driving}}} \geq 1.5$$

Bearing capacity: Check using Terzaghi or Meyerhof methods; account for eccentric loading.

4.2 Basal Heave in Soft Clays (Terzaghi)

Factor of safety against basal heave:

$$FS = \frac{N_c \cdot S_u}{\gamma H + q}$$

Where:

- N_c = bearing capacity factor (depends on excavation width/depth ratio)
- S_u = undrained shear strength
- γ = total unit weight
- H = excavation depth
- q = surcharge

Required FS: 1.5–2.0 for temporary excavations; 2.0–3.0 for permanent .

If FS < 1.5: Sheeting or soldier piles must be carried below the base of cut to ensure stability .

4.3 Piping, Boiling, and Quick Conditions

Piping occurs if the seepage exit gradient equals about 1.0 [50] :

$$i_{\text{exit}} = \frac{h}{d_2} \times \frac{\phi_2}{\phi_1 + \phi_2}$$

Heave occurs if uplift force at sheeting toe exceeds submerged weight of overlying soil column [50] .

Safety factor against piping/heave: 1.5 to 2.0 required [50] .

Prevention: Sufficient sheeting penetration below excavation base; filters; relief wells.

4.4 Ground Settlement and Adjacent Structures

Preliminary settlement estimate per AASHTO Figure C11.9.3.1-1. **Excludes** settlement from heavy construction surcharge, dewatering, foundation settlement, or poor construction practice [41] .

Mitigation: Underpinning, ground improvement, reduced surcharge, or staged excavation.

4.5 Global Stability

Check slope stability through the wall and behind the excavation. Failure surfaces passing just behind anchors or deadmen must be analyzed [41] .

4.6 Common Errors – Stability

Error	Correct Approach
Using FS = 1.0 for basal heave	FS ≥ 1.5 (temporary); ≥ 2.0 (permanent)
Forgetting piping check	Check exit gradient; ensure FS 1.5–2.0
Neglecting global stability	Always check slope stability through and behind wall
Ignoring adjacent structure settlement	Estimate settlement; mitigate if excessive

5. Groundwater Control and Dewatering

5.1 Dewatering Methods

Method	Description	Application
Wellpoints	Small diameter wells at close spacing; vacuum pumping	Shallow to medium depth (≤ 15–20 ft); suitable where submergence is small
Deep wells	Large diameter wells with submersible pumps	Deep excavations (up to 90 m)
Eductors (jet educators)	High-pressure water jet creates vacuum	Low permeability soils; deep drawdown
Sump pumping	Direct pumping from sumps in excavation	Supplementary; not primary dewatering

5.2 Grouting and Cutoff Walls

- **Grout curtains:** Reduce seepage; used where dewatering alone is insufficient
- **Cutoff walls:** Slurry walls or sheet piles extending to impermeable layer

5.3 Filters and Drainage Systems

- **Filter design (Terzaghi):** $D_{15,\text{filter}} / D_{85,\text{soil}} < 4\text{–}5$
- **Drainage blankets:** Relieve hydrostatic pressure behind walls
- **Weep holes:** Prevent buildup of water pressure

5.4 Effect on Adjacent Structures

Dewatering can cause **ground settlement** due to:

- Increased effective stress from lowered water table
- Loss of fines through pumping
- Compaction of loose sands

Mitigation: Recharge wells, cut-off walls, controlled pumping rates.

5.5 Common Errors – Groundwater

Error	Correct Approach
Using sump pumping as primary dewatering	Not effective for deep excavations; use wellpoints or deep wells
Forgetting filter design	Filter must prevent soil loss while allowing water flow
Neglecting settlement from dewatering	Estimate settlement; monitor adjacent structures
Using wellpoints for deep excavations	Wellpoints limited by suction lift (~15–20 ft)

6. Construction Sequence and Monitoring

6.1 Excavation Sequence

1. Install wall elements (soldier piles, sheet piles, or diaphragm wall)
2. Excavate to first strut/anchor level
3. Install strut/anchor; pre-load if required
4. Excavate to next level; repeat
5. Continue to final excavation depth
6. Install base slab or foundation
7. Remove struts/anchors as permanent structure provides support

Key principle: Excavate in stages; install support immediately after each stage.

6.2 Pre-Loading and Prestressing

- **Struts:** Pre-load to **50–80%** of design load to reduce wall movement
- **Anchors:** Lock-off at **75–100%** of design load per FHWA GEC-4

6.3 Monitoring Instrumentation

Instrument	Measures	Purpose
Piezometers	Groundwater pressure	Verify dewatering effectiveness
Inclinometers	Lateral wall movement	Detect excessive deflection
Settlement points	Vertical ground movement	Monitor adjacent structures
Load cells	Strut/anchor load	Verify design assumptions
Strain gauges	Wall bending	Validate structural design

6.4 Observational Method

1. **Predict** behavior (design)
2. **Monitor** actual behavior
3. **Compare** prediction vs. observation
4. **Adjust** design/construction if needed

Contingency planning: Pre-defined action levels for monitoring data.

6.5 Common Errors – Monitoring

Error	Correct Approach
Installing inclinometers after excavation	Install before excavation to establish baseline
Not pre-loading struts	Pre-load to reduce wall movement
Ignoring monitoring data	Use observational method; act on deviations
No contingency plan	Define action levels before construction

7. OSHA and Code Requirements

7.1 OSHA 1926 Subpart P – Excavations

Soil Classification (Appendix A):

Type	Description	Max Slope (H:V)	Max Depth (ft)
Stable Rock	Natural solid mineral matter	Vertical	Unlimited
Type A	Cohesive; unconfined compressive strength ≥ 1.5 tsf	3/4:1	20
Type B	Cohesive; 0.5–1.5 tsf; or granular with cohesion	1:1	20
Type C	Granular; < 0.5 tsf; or submerged/flowing	1.5:1	20

Key provisions:

- Protective systems required for excavations **5 ft or deeper**
- **Written design by a PE** required for excavations **20 ft or deeper** (or when tabulated data is exceeded)
- **Daily inspections** required before work and after events (rain, vibration, etc.)
- **Access/egress** required within **25 ft** of lateral travel for employees in trenches 4 ft or deeper

7.2 Written Design Requirement

OSHA 1926.652(b)(4)(i): Design by a registered professional engineer required for:

- Excavations **20 ft or deeper**
- Sloping/benching systems not using Option 1 or 2
- Any protective system not using tabulated data

7.3 Inspection Requirements

Inspection Type	Frequency
Daily	Before work begins
After events	Rain, vibration, equipment impact
By competent person	Trained in soil classification and protective systems

7.4 Access and Egress

- **Ladders** required for trenches **4 ft or deeper**
- Within **25 ft** of lateral travel
- **Structural ramps** designed by competent person

7.5 Common Errors – OSHA

Error	Correct Approach
Forgetting written design for excavations > 20 ft	PE-stamped drawings required
Skipping daily inspections	Required before work each day
No access within 25 ft	Ladders required for trenches ≥ 4 ft
Misclassifying soil	Type A, B, C per Appendix A criteria

8. Practice Problems with Detailed Solutions

Problem 1 – Active Earth Pressure

Problem: A 20 ft deep excavation in sand with $\phi' = 30^\circ$, $\gamma = 120$ pcf, and no groundwater. Calculate the total active earth pressure per foot of wall.

Solution:

1. $K_a = \tan^2(45^\circ - 30^\circ/2) = \tan^2(30^\circ) = 0.333$
2. $P_a = \frac{1}{2}\gamma H^2 K_a = 0.5 \times 120 \times 20^2 \times 0.333$
3. $P_a = 0.5 \times 120 \times 400 \times 0.333 = 8,000$ lb/ft
4. **Answer: 8,000 lb/ft**

Problem 2 – Basal Heave

Problem: A 25 ft deep excavation in soft clay with $S_u = 500$ psf, $\gamma = 110$ pcf, and $N_c = 7.0$. No surcharge. Check basal heave stability.

Solution:

1. $FS = \frac{N_c S_u}{\gamma H + q} = \frac{7.0 \times 500}{110 \times 25 + 0}$
2. $FS = \frac{3{,}500}{2{,}750} = 1.27$
3. Required FS = 1.5–2.0 for temporary excavation
4. **FS = 1.27 < 1.5 → Not adequate; sheeting must penetrate below base of cut**

Problem 3 – Piping Safety Factor

Problem: A sheet pile wall is driven 10 ft below the excavation base. The seepage exit gradient $i_{\text{exit}} = 0.45$. What is the FS against piping?

Solution:

1. Critical gradient $i_c \approx 1.0$
2. $FS = i_c / i_{\text{exit}} = 1.0 / 0.45 = 2.22$
3. Required FS = 1.5–2.0
4. **FS = 2.22 > 2.0 → Adequate** [50]

Problem 4 – Strut Load

Problem: A braced excavation has struts at 10 ft horizontal spacing. The apparent earth pressure at the strut level is 1,200 psf. What is the strut load?

Solution:

1. Tributary area per strut = 10 ft × 10 ft (vertical spacing) = 100 ft²
2. Strut load = 1,200 psf × 100 ft² = 120,000 lb = 120 kips
3. **Answer: 120 kips**

Problem 5 – Anchor Bond Length

Problem: A tieback anchor must extend beyond the active failure wedge. The active wedge extends 15 ft behind the wall face. The anchor is inclined at 15° from horizontal. What is the minimum unbonded length?

Solution:

1. Minimum unbonded length (measured along anchor) = 15 ft / cos(15°) = 15.5 ft
2. Also check minimum per FHWA: 3 m (9.8 ft) for bar, 4 m (13.1 ft) for strand
3. 15.5 ft > 13.1 ft → **15.5 ft governs**
4. **Answer: 15.5 ft**

Problem 6 – OSHA Written Design

Problem: A contractor proposes a 22 ft deep excavation using timber shoring. Is a written design by a PE required?

Solution:

1. Excavation depth = 22 ft > 20 ft
2. OSHA 1926.652(b)(4)(i) requires design by a registered PE for excavations **20 ft or deeper**
3. **Answer: Yes, written design by a PE is required**

9. High-Yield Checklist

Earth Pressures

□ $K_a = \tan^2(45^\circ - \phi/2)$

□ $K_p = \tan^2(45^\circ + \phi/2)$

□ $K_0 = 1 - \sin\phi$ (normally consolidated)

□ Water pressure $P_w = 62.4 \times h_w$

□ Surcharge $P_s = qK_a$

Design

□ Soldier piles: effective width up to 3× hole diameter (arching)

□ Sheet piles: cantilever limited to ~18 ft

□ Struts: max 67% failure load (temporary); 50% (permanent)

□ Anchors: unbonded length ≥ 3 m (bar) or 4 m (strand); bond length ≤ 12 m

□ Buckling: check $KL/r \leq 200$

Stability

□ Basal heave: FS ≥ 1.5–2.0

□ Piping: FS ≥ 1.5–2.0

□ Overturning: FS ≥ 1.5

□ Sliding: FS ≥ 1.5

Groundwater

□ Wellpoints: shallow (≤ 15–20 ft)

□ Deep wells: deep excavations

□ Filters: prevent soil loss

□ Dewatering settlement: monitor adjacent structures

OSHA

□ Protective systems required ≥ 5 ft

□ PE design required ≥ 20 ft

□ Daily inspections

□ Access within 25 ft for trenches ≥ 4 ft

Monitoring

□ Piezometers: water pressure

□ Inclinometers: wall movement

□ Settlement points: adjacent structures

□ Load cells: strut/anchor loads

10. Self-Test

1. What is the Rankine active earth pressure coefficient for $\phi' = 30°$?

- A) 0.25
- B) 0.333
- C) 0.50
- D) 3.0

Answer: B) 0.333

2. At what depth must a PE-stamped written design be provided for an excavation?

- A) 5 ft
- B) 10 ft
- C) 20 ft
- D) 30 ft

Answer: C) 20 ft

3. What is the maximum bonded length for a ground anchor per FHWA?

- A) 6 m
- B) 9 m
- C) 12 m
- D) 15 m

Answer: C) 12 m

4. What is the required FS against basal heave for a temporary excavation?

- A) 1.0
- B) 1.25
- C) 1.5–2.0
- D) 3.0

Answer: C) 1.5–2.0

5. Which dewatering method is limited by suction lift to about 15–20 ft?

- A) Deep wells
- B) Wellpoints
- C) Eductors
- D) Sump pumping

Answer: B) Wellpoints

6. What is the maximum allowable working load of a strut in a temporary excavation as a percentage of mean failure load?

- A) 50%
- B) 67%
- C) 75%
- D) 100%

Answer: B) 67%

7. What soil type has an unconfined compressive strength of 0.5–1.5 tsf?

- A) Type A
- B) Type B
- C) Type C
- D) Stable Rock

Answer: B) Type B

8. What is the minimum unbonded length for a strand anchor per FHWA?

- A) 2 m
- B) 3 m
- C) 4 m
- D) 6 m

Answer: C) 4 m

9. Which instrument measures lateral wall movement?

- A) Piezometer
- B) Inclinometer
- C) Load cell
- D) Settlement point

Answer: B) Inclinometer

10. What is the minimum surcharge load for construction equipment per typical specifications?

- A) 50 psf
- B) 100 psf
- C) 150 psf
- D) 200 psf

Answer: B) 100 psf

11. Flagged Items – Outside NCEES Scope or Requiring Verification

The following items could not be fully verified against NCEES-published specifications or the PE Civil Reference Handbook, Version 2.0:

1. **Exact FS values for temporary vs. permanent excavations:** The 1.5–2.0 range for temporary and 2.0–3.0 for permanent is from general geotechnical practice . NCEES may use different values. Verify against the PE Civil Reference Handbook.
2. **Peck empirical pressure diagrams:** The empirical pressure distributions (P1 = γH − 4Su, P2 = 0.2γH, etc.) are from Peck (1969) [31] . The exam may or may not require these exact formulas.
3. **SI unit equivalents:** All formulas are presented in USCS. SI conversions should be verified against the PE Civil Reference Handbook.
4. **Mononobe-Okabe seismic earth pressure:** The exact formulation for seismic earth pressure is complex and may be outside the typical exam scope. Verify if NCEES includes seismic SOE design.
5. **FHWA GEC-4 corrosion protection classes:** The Class I, II, III corrosion protection systems are from FHWA GEC-4 . The exam may test conceptual understanding rather than exact class requirements.

6. **OSHA soil classification tests:** The visual and manual tests for soil classification (Appendix A) are detailed in the OSHA regulations [70] . The exam may test general classification criteria rather than specific test procedures.
7. **Dewatering system design calculations:** The wellpoint and deep well design equations (flow to wells, radius of influence) are complex and may be outside the typical exam scope. Verify against the PE Civil Reference Handbook.

This review is designed to maximize readiness for the NCEES PE Civil: Construction exam. It does not guarantee a passing score. Always verify formulas and provisions against the current PE Civil Reference Handbook and referenced standards during the exam.

78 web pages

Act as a licensed PE an...
TXT 6.01KB

Thought for 5 seconds

The user wants a comprehensive exam-focused review of Construction Loads on Permanent Structures for the PE Civil Construction exam, including erection analysis and equipment loading. I need to search for relevant information on ASCE 37 load combinations, construction live loads, crane loads on structures, erection analysis, OSHA Subpart CC, and dynamic impact factors. I'll conduct multiple searches to gather this information.

Found 55 web pages

The search results provide some useful links. I'll open the ASCE 37 standard page, the construction live load article, the crane outrigger load distribution discussion, the AISC temporary supports page, the OSHA Subpart CC page, and the dynamic impact factor article.

Read 6 pages Design Loads on Structures during Construction | Books ↗ Construction Live Load! ↗ The service road will be made out of multiple prefabricated prestressed concrete slabs with a centre- to- centre ... ↗ conference preview ↗ 1926 Subpart CC - Cranes and Derricks in Construction ↗ View All

The opened pages provide some information. The ASCE 37 page gives an overview. The Eng-Tips thread mentions ASCE 37-02 live load levels (20, 25, 50, 75 psf). The TU Delft page discusses crane outrigger loads on concrete slabs. The AISC conference preview discusses erection engineering. The OSHA page is just a landing page. The OUCI page discusses dynamic loads from concrete pumps. I need more specific information on load combinations, crane load distribution, and erection analysis. I'll search for more specific queries.

Found 56 web pages

> The search results provide some relevant information. I'll open some of the more promising links to gather more details.

Read 4 pages Design Loads on Structures during Construction (37-14R2019) ↗
AISC Design Guide 10 "Erection Bracing of Low-Rise Structural Steel Buildings"; Its Development and Practical Us... ↗
S. K. Ghosh Associates LLC ↗ Investigating Moving and Lifting Impact Factors of a Crane Runway Girder ↗

> The search results provide some relevant information. I'll now organize the review. The user wants a comprehensive exam-focused review. I'll structure it with sections covering types of construction loads, load combinations, erection analysis, equipment loading, analysis methods, safety and code requirements, and common errors. I'll include formulas, worked examples, and practice problems. I'll flag items that may require verification.

Construction Loads on Permanent Structures for the NCEES PE Civil: Construction Exam

Exam-Focused Review | Design for Support of Construction Loads

Scope Note: This review covers only **Construction Loads on Permanent Structures**, including erection analysis and equipment loading, within the "Design for Support of Construction Loads" area. Formwork, falsework, shoring, scaffolding, and excavation support are referenced only where necessary to explain construction loads on permanent structures. All content is aligned with the NCEES PE Civil: Construction CBT Exam Specifications and the PE Civil Reference Handbook, Version 2.0.

Important: This topic is distinct from temporary structure design. The focus is on evaluating whether the **permanent structure** can safely support loads imposed **during construction**, before it reaches its final in-service condition.

1. Types of Construction Loads

1.1 Core Definitions

Load Type	Definition	Typical Source
Dead load of partially completed structure	Weight of incomplete framing, slabs, walls, and other permanent elements present during construction	Self-weight of steel, concrete, masonry
Construction live load	Temporary loads from workers, equipment, material storage, and temporary facilities	Personnel, tools, stored materials
Equipment loading	Loads from cranes, concrete pumps, delivery vehicles, aerial platforms	Crane outriggers, pump trucks, concrete buggies
Erection loads	Forces from lifting operations, temporary supports, guying, and bracing	Crane pick points, temporary guys, falsework
Wind loads during construction	Lateral loads on partially completed structure	ASCE 37 reduced wind speeds
Seismic loads during construction	Reduced seismic forces for short exposure duration	ASCE 37
Impact and dynamic loads	Amplified loads from equipment operation and material handling	Crane hoisting, concrete pumping
Thermal loads	Environmental temperature effects during construction	Daily temperature cycles

1.2 ASCE 37 Construction Live Loads

ASCE 37-14 provides construction live load levels based on the "degree" of construction activity [11] :

Degree of Construction	Uniform Live Load (psf)	Typical Use
Very light duty	20	Sparsely populated; small amounts of materials
Light duty	25	General construction with light materials
Medium duty	50	Standard construction with moderate material storage
Heavy duty	75	Heavy material storage, equipment staging

Critical exam point: These values apply to the **permanent structure during construction**, not to formwork or temporary structures. ACI 347R specifies a **minimum combined dead + live load of 100 psf** for formwork, or **125 psf** when using buggy transport .

1.3 Equipment Loading

Equipment loads are typically applied as **concentrated loads** at points of contact. Key considerations:

- **Crane outriggers:** Apply concentrated loads to slabs or foundations
- **Concrete pumps:** Dynamic loads requiring amplification factors
- **Delivery vehicles:** Wheel loads distributed through slabs-on-grade
- **Aerial platforms:** Concentrated loads at outrigger positions

1.4 Common Errors and Exam Traps – Load Types

Error	Correct Approach
Using in-service live loads for construction	Use ASCE 37 construction live loads (20–75 psf)
Forgetting dead load of incomplete structure	Include partially completed framing and slabs
Neglecting dynamic amplification	Apply impact factors to equipment loads

Error	Correct Approach
Confusing formwork live loads with construction live loads	Formwork: 50 psf minimum; Construction: 20–75 psf per ASCE 37

2. Load Combinations and Design Criteria

2.1 ASCE 37 Load Combinations

ASCE 37-14 provides load combinations suitable for either **strength design (LRFD/USD)** or **allowable stress design (ASD)** [1] . The standard addresses partially completed structures and temporary support structures [3] .

Key ASD load combinations for construction:

$$D + L_{\text{construction}}$$

$$D + 0.75L_{\text{construction}} + 0.75W$$

$$D + 0.75L_{\text{construction}} + 0.75S$$

Where:

- D = dead load of partially completed structure
- $L_{\text{construction}}$ = construction live load (20–75 psf)
- W = wind load (reduced per ASCE 37)
- S = snow load

Critical exam point: ASCE 37 permits **less stringent loads and load factors** for areas designated for construction activity and not subjected to design occupancies [link] .

2.2 Reduced Wind Loads During Construction

ASCE 37 incorporates provisions for adjusting wind loads for short-term exposure during construction for up to five years [8][9] . Design wind speed reduction factors [8][9] :

Construction Period	Wind Speed Reduction Factor
Less than six weeks	0.75
Six weeks to one year	0.80
One to two years	0.85
Two to five years	0.90

Example: For a structure under construction for less than six weeks, the wind load is **56% (0.75 × 0.75)** of that applied to a permanent structure, due to the reduced probability of exposure to the design wind event [89] .

2.3 Reduced Seismic Loads During Construction

ASCE 37 permits reduced seismic design forces for construction conditions due to the short exposure duration. The reduction is based on the probability of a seismic event occurring during the construction period.

2.4 Strength vs. Serviceability During Construction

Design Approach	When to Use	Notes
ASD	Temporary conditions, simple structures	Use service loads with allowable stresses
LRFD/USD	When required by specification or for critical structures	Use ASCE 37 load factors
Serviceability	Deflection and camber checks	Construction deflections may exceed in-service limits

2.5 Common Errors and Exam Traps – Load Combinations

Error	Correct Approach
Using in-service load combinations for construction	Use ASCE 37 combinations (reduced wind, seismic, live)
Forgetting wind speed reduction factors	Apply 0.75–0.90 based on construction duration
Using full seismic loads during construction	Reduce per ASCE 37
Applying LRFD factors to ASD design	Match load factors to design method

3. Erection Analysis

3.1 Stability of Partially Erected Structures

The **erector** is responsible for the stability of the steel frame through all stages of construction until the final structure is complete [28] . Key requirements from AISC:

- Section M4.2 of the AISC Specification: "The structure shall be secured to support dead, erection and other loads anticipated to occur during the period of erection" [28]
- Section 7.10.3 of the AISC Code of Standard Practice: "The erector shall determine, furnish and install all temporary supports, such as temporary guys, beams, falsework, cribbing or other elements required" [28]

3.2 Temporary Supports, Guying, and Bracing

When temporary bracing is required:

- Steel frames that are stable only in their completed form
- Structures where the completed diaphragm (deck, concrete slab) is not yet present
- Frames with insufficient inherent stability during erection

Design requirements:

- Temporary bracing shall be verified to have adequate **strength and stiffness** to provide stability

- Bracing must resist "loads comparable in intensity to those for which the structure was designed"
- Temporary supports are designed, furnished, and installed by the **contractor/erector**

AISC Design Guide 10 provides guidance for the design of temporary lateral support systems and components for low-rise steel buildings [33] .

3.3 Erection Sequence Effects

Erection sequence affects:

- **Member forces** — partially completed frames have different force distributions than completed frames
- **Deflections** — temporary loads may produce deflections that affect fit-up
- **Stability** — intermediate stages may be less stable than the completed structure

3.4 P-Delta and Second-Order Effects

During erection, the structure may be more flexible than in its completed state. P-delta effects must be considered when:

- The structure is partially braced
- Large axial loads act on flexible columns
- Story drift is significant during construction

3.5 Crane Pick Points and Load Distribution

Crane picks impose concentrated loads on the permanent structure. Key considerations:

- **Pick point locations** must be analyzed for local strength
- **Load distribution** through the structure to the foundation
- **Impact factors** for dynamic effects during lifting

3.6 Common Errors and Exam Traps – Erection Analysis

Error	Correct Approach
Assuming the structure is stable at all erection stages	Analyze each intermediate stage for stability
Forgetting temporary bracing	Required when structure is stable only in completed form
Neglecting P-delta effects during erection	Check second-order effects for flexible stages
Ignoring crane pick point loads	Analyze concentrated loads at pick points

4. Equipment Loading on Permanent Structures

4.1 Crane Loads on Slabs, Beams, and Foundations

Crane outrigger loads are applied as **concentrated point loads**. The load distribution depends on:

- Crane configuration (outrigger positions, base dimensions)
- Load being lifted
- Dynamic effects during hoisting

Outrigger load distribution on concrete slabs:

- Load spreads through the slab at a **1:1 gradient** through the depth (in both directions)
- Mats or dunnage may be used to distribute loads over a larger area
- The subgrade must be checked for bearing capacity at the larger footprint

4.2 Outrigger and Track Loading on Permanent Slabs

Load Type	Application	Key Consideration
Outrigger pads	Concentrated point loads	Check punching shear and flexure in slab
Track loads	Distributed line loads	Check slab capacity over track length

Load Type	Application	Key Consideration
Mats/dunnage	Distributed loads	Reduce bearing pressure; check mat thickness

4.3 Concrete Pump and Delivery Vehicle Loads

Concrete pumps impose:

- **Static loads** — weight of pump truck and concrete in boom
- **Dynamic loads** — from boom operation and concrete flow
- **Amplification factor** — dynamic effects require multiplying static loads by an amplification factor [49]

4.4 Dynamic and Impact Factors for Equipment

Equipment	Dynamic Factor	Source
Tower crane, portal crane, mobile crane	1.3	General practice
Crane vertical wheel load	25% impact	IS 875
Crane lateral surge	5% of wheel load	IS 875
Lifting load (hoisting)	> moving load	Research confirms [96]

4.5 Load Distribution and Tributary Areas

Equipment loads must be distributed to structural members through:

1. **Direct contact** — outrigger pad on slab
2. **Punching shear** — check slab capacity at concentrated load
3. **Load spreading** — through slab depth at 1:1 gradient
4. **Tributary area** — to beams, columns, and foundation

4.6 Common Errors and Exam Traps – Equipment Loading

Error	Correct Approach
Neglecting dynamic and impact loads	Apply appropriate impact factors (1.3 for cranes)
Ignoring load spreading through slab depth	Distribute at 1:1 gradient
Forgetting punching shear check	Check slab capacity at concentrated loads
Using static loads only for concrete pumps	Apply dynamic amplification factor

5. Analysis Methods for Construction Loads

5.1 Structural Analysis of Permanent Structures

The permanent structure must be analyzed under construction loads using:

- **Influence lines and moving loads** for erection equipment
- **Beam, column, and connection checks** during construction
- **Deflection and camber considerations** during erection

5.2 Influence Lines and Moving Loads

For erection equipment (cranes, delivery vehicles) that move across the structure:

- Construct **influence lines** for critical response quantities (moment, shear, deflection)
- Apply **moving load patterns** to maximize the response
- Consider **dynamic effects** from moving loads

5.3 Deflection and Camber Considerations

Consideration	Requirement
Erection deflections	May exceed in-service limits; must not impair fit-up
Camber	May be required to compensate for dead load deflections

Consideration	Requirement
Construction sequence	Affects final geometry

5.4 Buckling and Stability of Partially Erected Members

Compression members during erection:

- May have **larger effective lengths** (less bracing)
- May be subjected to **eccentric loads** from temporary supports
- Must be checked for **buckling** with appropriate K factors

5.5 Foundation and Bearing Capacity Checks

Construction loads may exceed in-service loads at certain foundation locations. Check:

- **Bearing capacity** under construction loads
- **Settlement** during construction
- **Global stability** of temporary conditions

5.6 Common Errors and Exam Traps – Analysis Methods

Error	Correct Approach
Using in-service analysis models for construction	Create separate construction-stage models
Forgetting deflection checks during erection	Check deflections for fit-up and serviceability
Neglecting foundation capacity under construction loads	Check bearing and settlement
Using in-service K factors for erection stages	Use appropriate effective lengths

6. Safety and Code Requirements

6.1 ASCE 37 Requirements

- Minimum design requirements for construction loads, load combinations, and load factors [1]
- Applicable to partially completed structures and temporary support structures [3]
- Loads suitable for strength design (LRFD/USD) or ASD [1]

6.2 OSHA 1926 Subpart Q – Concrete and Masonry Construction

- Formwork must support all anticipated vertical and lateral loads [28]
- Drawings/plans available at jobsite
- Shoring inspection before, during, and after placement

6.3 OSHA 1926 Subpart CC – Cranes and Derricks in Construction

OSHA 1926 Subpart CC governs crane operations and requires [44] :

- Compliance with manufacturer load charts
- Operator certification
- Ground/surface suitability assessment
- Site-specific lift plans for critical lifts

6.4 AISC Code of Standard Practice

Section 7.10.3: Erector shall determine, furnish, and install all temporary supports required [28] .

6.5 Written Design by a Qualified Person

Written design by a qualified person is required for:

- Critical erection operations
- Temporary supports for structures stable only in completed form
- Tiered single post shores (OSHA 1926.703(b)(8))

6.6 Common Errors and Exam Traps – Safety and Code

Error	Correct Approach
Missing written design for critical erection	Required by AISC and OSHA
Not checking ground/surface for crane loads	Required by OSHA 1926 Subpart CC
Using in-service load combinations	Use ASCE 37 construction combinations
Forgetting contractor responsibility for temporary supports	AISC Code of Standard Practice assigns to erector

7. Worked Examples and Practice Problems

Problem 1 – Construction Live Load Selection

Problem: A steel-framed building is under construction. Workers are erecting steel with moderate amounts of material stored on the deck. What construction live load should be used per ASCE 37?

Solution:

1. Moderate material storage = **Medium duty**
2. ASCE 37 medium duty = **50 psf**
3. **Answer: 50 psf**

Problem 2 – Wind Speed Reduction

Problem: A structure will be under construction for 4 months (approximately 17 weeks). What wind speed reduction factor applies per ASCE 37?

Solution:

1. 4 months = approximately 17 weeks
2. Six weeks to one year → **0.80**
3. Wind pressure reduces by 0.80^2 = **0.64** (64% of permanent design pressure)
4. **Answer: Factor = 0.80; pressure = 64% of permanent**

Problem 3 – Crane Outrigger Load Spreading

Problem: A crane outrigger applies a 50,000 lb load to a concrete slab. The outrigger pad is 2 ft × 2 ft. The slab is 8 in. thick. What is the bearing pressure at the slab-subgrade interface assuming 1:1 load spreading?

Solution:

1. Pad area = 2 × 2 = 4 ft²
2. Load spreads at 1:1 through 8 in. (0.67 ft) slab depth
3. Spread dimensions = (2 + 2×0.67) × (2 + 2×0.67) = 3.33 × 3.33 = 11.1 ft²
4. Bearing pressure = 50,000 / 11.1 = **4,500 psf**
5. **Answer: 4,500 psf** (check against subgrade capacity)

Problem 4 – Dynamic Impact Factor

Problem: A mobile crane will lift a 20,000 lb load. What is the design load including dynamic impact?

Solution:

1. Dynamic factor for mobile crane = **1.3**
2. Design load = 20,000 × 1.3 = **26,000 lb**
3. **Answer: 26,000 lb**

Problem 5 – Construction Load Combination

Problem: A partially completed structure has D = 80 psf, L_construction = 50 psf, and W = 20 psf (reduced). Using ASD, what is the governing load combination?

Solution:

1. $D + L = 80 + 50 = 130$ psf
2. $D + 0.75L + 0.75W = 80 + 37.5 + 15 = 132.5$ psf
3. **Answer: 132.5 psf** (governs)

Problem 6 – Erection Stability

Problem: A steel frame is stable only after the concrete deck is placed. During erection, temporary guys are required. Who is responsible for designing and installing the temporary guys?

Solution:

1. Per AISC Code of Standard Practice Section 7.10.3: **The erector** determines, furnishes, and installs all temporary supports
2. **Answer: The erector (contractor)**

8. High-Yield Checklist

Load Types

□ ASCE 37 construction live loads: 20, 25, 50, 75 psf

□ Dead load of partially completed structure included

□ Equipment loads as concentrated loads

□ Dynamic impact factors applied (1.3 for cranes)

Load Combinations

□ ASD: D + L; D + 0.75L + 0.75W

□ Wind speed reduction: 0.75–0.90 by duration

□ Seismic reduction per ASCE 37

□ Match load factors to design method (ASD vs. LRFD)

Erection Analysis

□ Structure must be stable at all intermediate stages

□ Temporary bracing when stable only in completed form

□ Erector responsible for temporary supports

□ P-delta effects checked for flexible stages

Equipment Loading

□ Outrigger loads as concentrated point loads

□ Load spreads at 1:1 through slab depth

□ Punching shear check at concentrated loads

□ Dynamic factors: 1.3 for cranes; 25% impact for wheel loads

Analysis Methods

□ Influence lines for moving equipment loads

□ Separate construction-stage analysis models

□ Deflection and camber checks

□ Foundation capacity under construction loads

Safety and Code

□ ASCE 37 for construction load combinations

□ OSHA 1926 Subpart CC for crane operations

□ AISC Code of Standard Practice for erection

□ Written design for critical erection operations

9. Self-Test

1. What is the ASCE 37 construction live load for medium-duty construction?

- A) 20 psf
- B) 25 psf
- C) 50 psf
- D) 75 psf

Answer: C) 50 psf

2. What is the wind speed reduction factor for a construction period of 3 months?

- A) 0.75
- B) 0.80
- C) 0.85
- D) 0.90

Answer: B) 0.80 (six weeks to one year)

3. Who is responsible for designing and installing temporary erection bracing?

- A) The structural engineer of record
- B) The erector (contractor)
- C) The owner
- D) The architect

Answer: B) The erector (contractor) (per AISC Code of Standard Practice)

4. What dynamic factor is typically applied to mobile crane loads?

- A) 1.0
- B) 1.1
- C) 1.3
- D) 2.0

Answer: C) 1.3

5. How does a concentrated outrigger load spread through a concrete slab?

- A) At 1:2 gradient
- B) At 1:1 gradient
- C) Uniformly across the slab
- D) No spreading occurs

Answer: B) At 1:1 gradient

6. Which standard provides minimum design loads for structures during construction?

- A) ASCE 7

- B) ASCE 37
- C) ACI 318
- D) AISC 360

Answer: B) ASCE 37

7. What is the minimum combined dead + live load for formwork per ACI 347R?

- A) 50 psf
- B) 75 psf
- C) 100 psf
- D) 125 psf

Answer: C) 100 psf

8. What OSHA subpart governs cranes and derricks in construction?

- A) Subpart L
- B) Subpart P
- C) Subpart Q
- D) Subpart CC

Answer: D) Subpart CC

9. When is temporary bracing required during steel erection?

- A) Never
- B) Only for tall buildings
- C) When the structure is stable only in its completed form
- D) Always

Answer: C) When the structure is stable only in its completed form

10. What is the approximate wind load percentage for a structure under construction for less than six weeks compared to a permanent structure?

- A) 25%

- B) 56%
- C) 75%
- D) 100%

Answer: B) 56% (0.75 × 0.75)

10. Flagged Items – Outside NCEES Scope or Requiring Verification

The following items could not be fully verified against NCEES-published specifications or the PE Civil Reference Handbook, Version 2.0:

1. **Exact ASCE 37 load combinations:** The ASD combinations presented (D + L; D + 0.75L + 0.75W) are representative. The exact combinations should be verified against ASCE 37-14 Section 2.3.
2. **SI unit equivalents:** All values are presented in USCS. SI conversions (kPa, kN, mm) should be verified against the PE Civil Reference Handbook.
3. **OSHA 1926 Subpart CC specific provisions:** The subpart governs crane operations broadly. Specific load distribution requirements on permanent structures may be covered by other standards or specifications.
4. **Dynamic amplification factors:** The 1.3 factor for mobile cranes is from general practice (EN 1991-3). The specific factor required by NCEES may differ. Verify against the PE Civil Reference Handbook.
5. **ASCE 37 seismic reduction provisions:** The exact reduction methodology for seismic loads during construction should be verified against ASCE 37-14.
6. **AISC Design Guide 10 prescriptive bracing systems:** The design guide provides prescriptive systems for low-rise buildings. For high-rise or complex structures, project-specific erection engineering is required.

This review is designed to maximize readiness for the NCEES PE Civil: Construction exam. It does not guarantee a passing score. Always verify formulas and provisions against the current PE Civil Reference Handbook and referenced standards during the exam.

104 web pages